

13

Optimization

This chapter describes how to optimize simulation runs in Eclipse:

- [Optimization](#)
- [Gradient option](#)
- [Adjoint gradient screening](#)
- [Petro-elastic model](#)

Optimization

	Eclipse 100
x	Eclipse 300

This chapter describes the optimization facilities within Eclipse.

Several aspects of an engineering study involve the optimization of an Eclipse model. Increasingly, these optimization tasks are being incorporated into software programs, often using the simulator as a 'black box' function evaluation in an outer optimization loop. Optimization methods are readily divided into gradient and non-gradient methods. Methods that do not use gradients of the objective function with respect to model parameters, such as simulated annealing, are typically only applied to greatly simplified models, as they require a great many runs of the simulator. However, even methods that do use gradient information can be prohibitively expensive if the gradients are derived numerically, from multiple runs of the 'black box'.

Eclipse production profile gradients can be calculated much more efficiently from within the simulator itself (see "[Gradient option](#)"). The SimOpt interactive application successfully makes use of these Eclipse gradients in an external gradient-based optimization loop, as an aid in the history matching phase of a study.

Optimization capabilities have also been built into Eclipse 300, which use gradients of a general objective function rather than the production profiles. The gradients are calculated using the Adjoint technique, which has significant performance advantages when the number of parameters is large. Production constraints are handled using a Lagrangian formulation of the problem, which also provides an enormous speed up compared to treatments external to Eclipse.

The sections that follow give a technical description of the various aspects of the optimization workflow environment within Eclipse 300. This is followed by some guidelines on the use of the optimization feature, a tabulated list of optimization keywords, and subsequently some examples of their use in the Eclipse 300 reservoir optimization option.

Non-linear gradient-based optimization

Optimization structure

The central task of optimization is to maximize some objective function, J , by variation of a set of control parameters, $\underline{\lambda}$. An example optimization problem is that of reservoir optimization, where J can be simply the total oil produced and $\underline{\lambda}$ are the controlling well rates during a simulation run.

This optimization task is invariably iterative. For gradient-based optimization, this can be represented schematically as shown in the following figure.

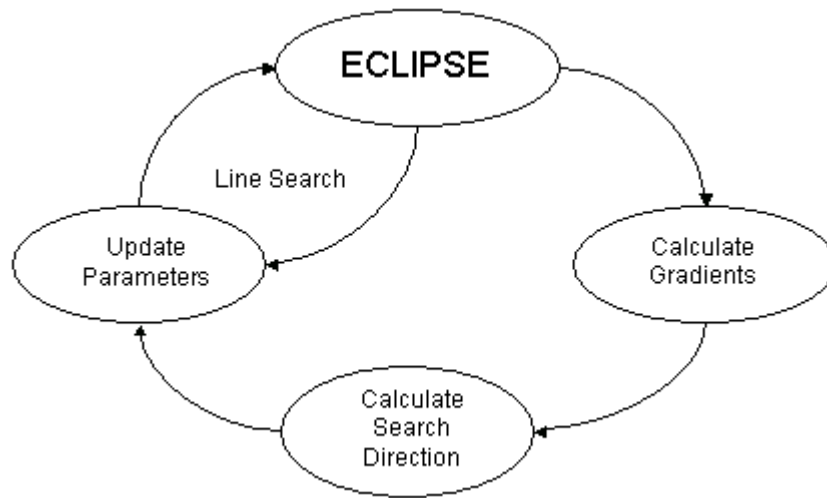


Figure 13.1. Schematic representation of gradient based optimization

This iterative structure is based on obtaining a search direction, s , from a set of gradients of the objective function with respect to the control parameters, $dJ/d\lambda$. A line search is performed using this initial search direction to find the optimum step size at which point the line search is terminated and a new direction calculated. This loop is repeated until one (or more) termination criteria are met. These termination criteria include lower limits on the relative changes of J or of λ .

This optimization loop structure has been implemented in Eclipse 300, and is controlled by keyword definitions given in the OPTIMIZE section of the Eclipse 300 data file.

Impact of constraints

The presence of constraints within an optimization greatly increases the complexity of the problem (Sokolnikoff and Redheffer, 1967). Constraints may vary from simple bounds on parameter values to full non-linear functional inequality relationships between parameters and Eclipse outputs, for example, the well and group production constraints.

The former are dealt with relatively easily, by a restriction of the step during a line search. However, the latter require modifications to the search direction.

Inequality constraints are most frequently found in the form $f(x, y) \leq K$, which may be rewritten as:

$$f(x, y) - K = V \leq 0 \quad \text{Eq. 13.1}$$

A constraint of this form is said to become “active” when the inequality becomes an equality, that is when $V = 0$. Otherwise the constraint is said to be “inactive”.

A simple example that illustrates the impact of constraints is shown in the following figure.

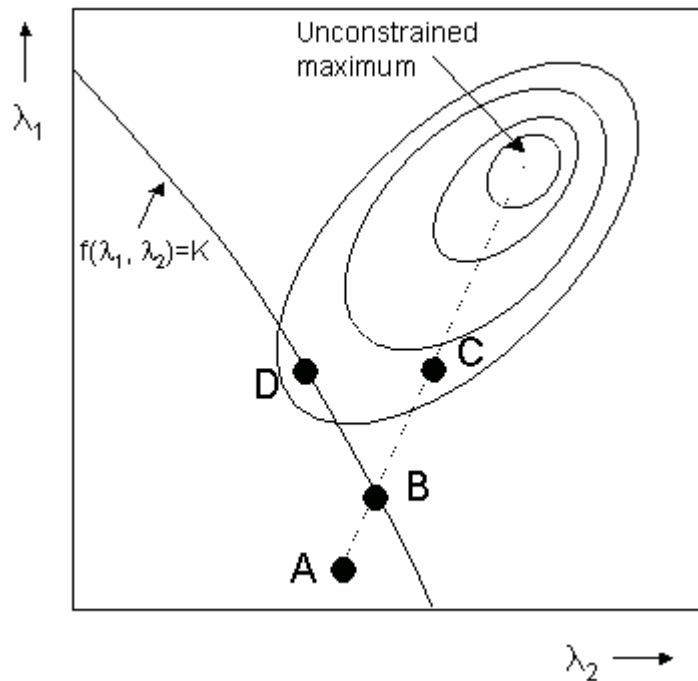


Figure 13.2. An Example of constrained optimization.

This diagram shows the objective function contour map for a problem with two variables, λ_1 and λ_2 , in which a nonlinear inequality constraint function $f(\lambda_1, \lambda_2) \leq K$ divides the parameter space into feasible (bottom left) and infeasible (top right) regions. From a starting position at A using a search direction indicated by the dashed line, a line search would have to terminate at B (to avoid violating the constraint). From B, an unconstrained search direction would point towards C, which violates the constraint. The direction we require is towards the constrained maximum at D.

The Lagrangian formulation is a well established mechanism for combining an objective function with equality constraints. For the example above, we form the following Lagrangian function: (Sokolnikoff and Redheffer, 1967)

$$L(\underline{\lambda}) = J(\underline{\lambda}) - \Phi V(\underline{\lambda}) \quad \text{Eq. 13.2}$$

For suitable values of the Lagrange multiplier, Φ , unconstrained optimization of the Lagrangian simultaneously achieves constrained optimization of the original objective function.

The Lagrangian formulation relies on being able to determine which inequality constraints are active, that is which are currently an equality constraint. As the above illustration indicated, a specialized line search algorithm can be used for just such a purpose. However, one potential problem of this approach is that it may dominate the optimization convergence - at worst we may end up following the Simplex algorithm of linear programming, with one iteration per constraint.

"Adjoint gradient calculations in Eclipse" shows how we use the Lagrangian formulation to construct the optimization gradients using the Adjoint method.

Adjoint gradient calculations

After a simulation run, the optimization workflow starts with the calculation of the gradients of the objective function with respect to the chosen parameters. These gradients must take into account the equality constraints imposed by the fluid flow calculations solved in Eclipse.

At each simulator timestep, Eclipse forces the fully implicit residual in each simulator grid block to be zero (see "Formulation of the Equations").

CAUTION: The adjoint gradient calculations require a fully implicit formulation of the equations. If you try to use the reservoir optimization feature without this type of formulation, the run will be aborted.

This cell residual constraint equation may be written as

$$R(X^{n+1}, X^n) = \frac{M(X^{n+1}) - M(X^n)}{dt} + F(X^{n+1}) + Q(X^{n+1}) = 0 \quad \text{Eq. 13.3}$$

where

the superscripts $n+1$ and n denote the current and previous timestep respectively, and

X^n are the solution variables (pressure and molar densities for example) at time n

$M(X^n)$ is the fluid-in-place in each simulator grid block at time n ,

$F(X^n)$ is the net flow rate into neighboring grid blocks at time n , and

$Q(X^n)$ is the net flow rate into wells at time n .

Note that the objective function is a function of both the control parameters $\underline{\lambda}$ and the simulation primary variables X ; that is $J = J(X, \underline{\lambda})$.

A Lagrangian formulation of the objective function is used in order to comply with the constraints imposed by the cell residuals,

$$L(X, \underline{\lambda}) = J(X, \underline{\lambda}) - \underline{\Phi} \cdot R(X, \underline{\lambda}) \quad \text{Eq. 13.4}$$

Note that when all the residuals are satisfied ($R = 0$) the Lagrangian is simply the objective function. The vector of Lagrange multipliers on the residuals, $\underline{\Phi}$, is known as the “adjoint vector”.

We are interested in forming the total parameter derivatives of this Lagrangian:

$$\frac{dL}{d\underline{\lambda}} = \frac{\partial L}{\partial \underline{\lambda}} + \frac{\partial L}{\partial X} \cdot \frac{\partial X}{\partial \underline{\lambda}} \quad \text{Eq. 13.5}$$

for which we require the partial Lagrangian derivatives:

$$\frac{\partial L}{\partial X} = \frac{\partial J}{\partial X} - \left(\underline{\Phi} \cdot \frac{\partial R}{\partial X} \right) \quad \text{Eq. 13.6}$$

and

$$\frac{\partial L}{\partial \underline{\lambda}} = \frac{\partial J}{\partial \underline{\lambda}} - \left(\underline{\Phi} \cdot \frac{\partial R}{\partial \underline{\lambda}} \right) \quad \text{Eq. 13.7}$$

as well as the solution derivatives $\frac{\partial X}{\partial \lambda}$.

Since $R = 0$, the vector of Lagrange multipliers can be chosen in any way.

As there is an implicit dependence of the solution variables on the control parameters, $X = X(\lambda)$, the natural choice of Φ is that which sets equation 13.6 to zero. This removes the need to calculate the solution derivatives. Hence we solve the adjoint set of linear equations:

$$\Phi \cdot \frac{\partial R}{\partial X} = \frac{\partial J}{\partial X} \quad \text{Eq. 13.8}$$

for the adjoint vector Φ . The dimension of the adjoint vector for each time period is that of the normal fluid flow Jacobian (number of grid cells times number of simulation variables).

As an example, scoping equation 13.8 into four distinct time periods, it may be written as:

$$[\Phi^1 \ \Phi^2 \ \Phi^3 \ \Phi^4] \begin{bmatrix} A^{11} & 0 & 0 & 0 \\ A^{21} & A^{22} & 0 & 0 \\ 0 & A^{32} & A^{33} & 0 \\ 0 & 0 & A^{43} & A^{44} \end{bmatrix} = \left[\frac{\partial J}{\partial X^1} \frac{\partial J}{\partial X^2} \frac{\partial J}{\partial X^3} \frac{\partial J}{\partial X^4} \right] \quad \text{Eq. 13.9}$$

where the A matrices are derivatives of the fluid flow residuals with respect to solution variables, given by

$$A^{mn} = \frac{\partial R^m}{\partial X^n} \quad \text{Eq. 13.10}$$

where m and n are time indices. By multiplying out $A^{mn} = \frac{\partial R^m}{\partial X^n}$, equation 13.9 term by term the general equation, which must be solved at any given time, index n is

$$\Phi^n A^{nn} = \frac{\partial J}{\partial X^n} - \Phi^{n+1} A^{n+1, n} \quad \text{Eq. 13.11}$$

This involves the value of the adjoint vector at time period, $n+1$. Since the only independent equation is the last one (for which we can solve for) then the adjoint Φ^4 equation, equation 13.11 must be solved backwards in time.

With the adjoint vector, Φ , chosen as the solution of this system, the Lagrangian gradients we require are:

$$\frac{dL}{d\lambda} = \frac{\partial J}{\partial \lambda} - \left(\Phi \cdot \frac{\partial R}{\partial \lambda} \right) \quad \text{Eq. 13.12}$$

Note: The effort of calculating these gradients is dominated by the solution for the adjoint vector, Φ , which is independent of the type or number of parameters. This makes the adjoint method very efficient for large number of parameters compared to other gradient calculation techniques.

Introducing Additional Constraints

In the reservoir simulator the fluid flow residual equations ($R = 0$) are constraints, which must always be observed as the objective function is optimized. Optionally, you may impose additional constraints on the system, for example limiting the production of a certain phase from a group of wells.

The Lagrangian approach allows us to readily incorporate these additional equality constraint equations, $V = 0$, by the introduction of further Lagrange multipliers, Ψ :

$$L(X, \underline{\lambda}) = J(X, \underline{\lambda}) - \Phi \cdot R(X, \underline{\lambda}) - \Psi \cdot V(X, \underline{\lambda}) \quad \text{Eq. 13.13}$$

The Lagrangian derivatives given in equations 13.5 to 13.7 above may be extended to include derivatives of the additional constraint functions. This allows us to form an extended adjoint system of equations, from which we obtain both $\underline{\Phi}$ and $\underline{\Psi}$. The Lagrangian gradient equations, 13.6 and 13.7, then become:

$$\frac{dL}{dX} = \frac{\partial J}{\partial X} - \left(\underline{\Phi} \cdot \frac{\partial R}{\partial X} \right) - \left(\underline{\Psi} \cdot \frac{\partial V}{\partial X} \right) \quad \text{Eq. 13.14}$$

$$\frac{dL}{d\underline{\lambda}} = \frac{\partial J}{\partial \underline{\lambda}} - \left(\underline{\Phi} \cdot \frac{\partial R}{\partial \underline{\lambda}} \right) - \left(\underline{\Psi} \cdot \frac{\partial V}{\partial \underline{\lambda}} \right) \quad \text{Eq. 13.15}$$

Notice that we have retained the special identity of the cell residual equations, as distinct from all other constraints. This allows us to solve for the Lagrange multipliers using modifications to the existing simulation fluid flow matrix A , rather than having to construct an augmented system.

In order to solve for $\underline{\Psi}$, we arbitrarily set $\frac{\partial L}{\partial \lambda_i} = 0$ in equation 13.15 for as many parameters, index i , as there are active constraints. Consequently, equation 13.15 only applies to those parameters not used in determining $\underline{\Psi}$. This effectively divides the parameter set into 'free' and 'dependent' subsets: $\underline{\lambda} = (\underline{\lambda}_f, \underline{\lambda}_d)$, where the free parameters are those for which we have Lagrangian gradients, and the dependent parameters are those which 'depend' on the free parameters using the constraint equations. This has important implications for the optimization workflow, as discussed in "Optimizer search direction".

Optimizer search direction

Given the Lagrangian gradients, $dL/d\underline{\lambda}$, the optimization workflow then proceeds by calculating a search direction, $\underline{s}$.

However, in the presence of equality constraints other than the cell residual equations, we do not have Lagrangian gradients for dependent parameters, $\underline{\lambda}_d$. The usual course of action is to determine the 'free' search direction, $\underline{s}_f$, update the free parameter values, $\underline{\lambda}_f$, and then invert the constraint equations to update $\underline{\lambda}_d$. Unfortunately, since the constraint equations are most generally non-linear, this inversion is iterative, and expensive, since each function evaluation requires a simulation run.

Instead, changes in the dependent parameters are determined from changes in the free parameters using a linearization of the perturbed constraint equations:

$$\frac{\partial R}{\partial X} \cdot \Delta X \frac{\partial R}{\partial \underline{\lambda}_d} \cdot \Delta \underline{\lambda}_d + \frac{\partial R}{\partial \underline{\lambda}_f} \cdot \Delta \underline{\lambda}_f = 0 \quad \text{Eq. 13.16}$$

$$\frac{\partial V}{\partial X} \cdot \Delta X \frac{\partial V}{\partial \underline{\lambda}_d} \cdot \Delta \underline{\lambda}_d + \frac{\partial V}{\partial \underline{\lambda}_f} \cdot \Delta \underline{\lambda}_f = 0 \quad \text{Eq. 13.17}$$

With $\Delta \underline{\lambda}_f = \underline{s}_f$, we solve this system for $\underline{s}_d = \Delta \underline{\lambda}_d$, thus obtaining the full search direction. In solving the above, the boundary conditions are that $\Delta X = 0$ at time zero, and so these equations must be solved forwards in time.

The modified workflow for determining the optimization search direction now consists of two distinct parts.

- First the search direction for the free parameters is determined during the backwards equation solution.
- Then the free parameter search direction is used to determine the search direction for the set of dependent parameters in a forward solution of the system.

Note: In this release, the calculation of the search direction from the gradients is performed using either a steepest descent or a conjugate gradient algorithm. The conjugate gradient algorithm requires slightly more computational work and storage. Due to the presence of the constraints, there is not much difference in the convergence rate of the two methods, although the conjugate gradient algorithm does appear to give a more stable convergence path.

Line search

Given a search direction, s , the standard line search algorithm is to determine some step size multiplier, α , which maximizes the objective function value. The control parameters are updated using the algorithm $\underline{\lambda} = \underline{\lambda} + \alpha \underline{s}$. The following figure illustrates the problem.

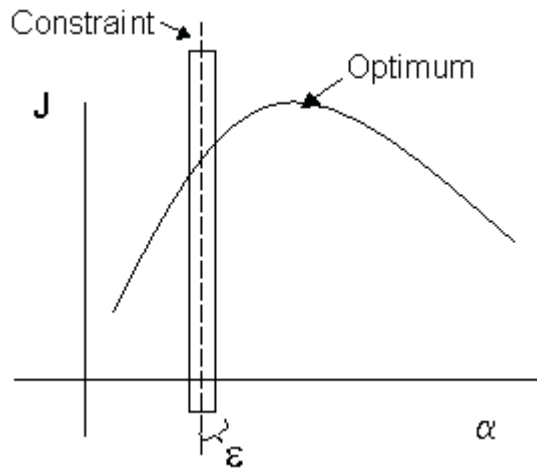


Figure 13.3. Line search

With no preferred scale, a line search may take many iterations to find the optimum position. However, along the way, we may encounter, or jump over, a new constraint. An indication of the position of the nearest constraint to the starting position of a line search can be obtained by expanding the constraint inequalities to give:

$$V_i(X + \alpha_i \Delta X, \underline{\lambda}_f + \alpha_i \Delta \underline{\lambda}_f) = V_i(X, \underline{\lambda}) + \alpha_i \left(\frac{\partial V_i}{\partial X} \cdot \Delta X \right) + \alpha_i \left(\frac{\partial V_i}{\partial \underline{\lambda}_f} \cdot \Delta \underline{\lambda}_f \right) \leq 0 \quad \text{Eq. 13.18}$$

Clearly, the closest constraint (that which would be first violated if an unrestricted amount of the search direction were used) is that for which the value of α is smallest when the inequality becomes an equality constraint. This may be obtained by rearranging equation 13.18, from which:

$$\alpha_i = -V_i / \left(\frac{\partial V_i}{\partial \underline{\lambda}_f} \cdot \Delta \underline{\lambda}_f + \frac{\partial V_i}{\partial X} \cdot \Delta X \right) \quad \text{Eq. 13.19}$$

This sets a useful guide for the line search algorithm in the presence of constraints, and in practice can save many iterations of the line search.

As a further aid to convergence of the overall optimization loop, the tolerance used in determining the proximity of a constraint, ϵ , can be specified to potentially 'catch' several new constraints at once.

Reservoir optimization

The Eclipse 300 reservoir optimization option allows you to optimize production by variation of well-based parameters. The option is triggered by the **RESOPT** keyword in the **RUNSPEC** section. This activates the processing of the **OPTIMIZE** section of a data set, in which you can specify particular forms of the objective function, parameters and constraints appropriate to the reservoir optimization task. These are described below.

The objective function

The reservoir optimization objective function is specified using the **OPTFUNC** keyword. This keyword permits specification of a very general function of the form:

$$J = \sum_i C_i \sum_{t_1(i)}^{t_2(i)} d_i(t) f_i(t) \Delta t \quad \text{Eq. 13.20}$$

where

$f_i(t)$ denotes a produced or injected fluid rate (for example, field oil production rate),

t_1 and t_2 are restart indexes defining the time range of interest,

C_i is a function-dependent weighting coefficient (whereby for example water production could be penalized), and d_i is a discount factor, given by:

$$d_i(t) = (1 + r_i)^{-N(t)} \quad \text{Eq. 13.21}$$

for annual discount rate r_i and simulation time $N(t)$ in fractional years.

The time limits appearing on the second summation allow functions to be included in the total objective function at different times with different weights, and different discount factors.

Control parameters

Parameters are specified using the **OPTPARS** keyword. In this release, the available parameters are:

- Well bottom hole pressures
- Well oil and gas production target rates
- Well oil, water and gas injection target rates
- The segment cross sectional area of a multisegment well

As an example, consider the bottom hole pressure at which a well is operated as a parameter. In general the value of this parameter varies over the course of a simulation, resulting in many control parameters for this well, one for each period of time. A general parameter is therefore denoted λ_p^t , where p identifies a specific control parameter (for example, bottom hole pressure for a particular well), and t scopes the parameter in time.

Parameters are created using the time axis defined by restart files. Thus, to vary well rates quarterly, restart files must be requested at three month intervals in the data set. For m user-specified parameters, and with the number of restart files defined as n , the complete vector of control parameters for the problem is thus:

$$\underline{\lambda} = \left[\left[\lambda_1^1 \lambda_1^2 \dots \lambda_1^n \right] \left[\lambda_2^1 \lambda_2^2 \dots \lambda_2^n \right] \cdot \dots \cdot \left[\lambda_m^1 \lambda_m^2 \dots \lambda_m^n \right] \right] \quad \text{Eq. 13.22}$$

This vector is reported in the `PRINT` file during an optimization run.

Note: In reservoir optimization, the total number of control parameters can be very high. For example, in a large study with perhaps 1000 wells under optimizer control, and with settings to be changed four times a year over a twenty-five year period, there are a total number of 10^5 control parameters. Thus an efficient gradient calculation scheme, such as the adjoint method, is critically important.

Production constraints

Production inequality constraints are specified using the `OPTCONS` keyword. The Reservoir Optimization feature allows you to specify constraints in the general form:

$$f(X, \underline{\lambda}) \leq K \quad \text{Eq. 13.23}$$

where

X are the reservoir solution variables,

$\underline{\lambda}$ is the vector of control parameters,

K is a constant, and

f is a fluid production or injection rate, from either a group or a well.

You may also specify the time period for which the constraint is to apply, defaulted to the entire simulation period.

File output

The reservoir optimization option writes output to the Eclipse `PRINT` file. For each iteration of the optimization procedure, the following are reported:

- Objective function value
- Parameter values
- Constraint values
- A list of constraints which are currently active
- Optimizer search direction
- Line search details.

As an option the results of each optimization iteration may be output to separate summary and restart files for visualization by Eclipse Office. This is achieved by using the `RPTOPT` keyword mnemonic `SEPARATE`.

Optimizer performance

Parameter selection

It is strongly recommended that, from the available selection of optimization parameters, you try the bottom hole pressure parameters first.

The reason for this is that production rates are insensitive to well target rates whenever a well falls onto bottom hole pressure control. With bottom hole pressures as parameters, the production rates are

only insensitive when a well is shut, which happens less frequently, and thus optimizations performed with such parameters are usually more successful.

Adjoint gradient accuracy

In general the most accurate gradients are obtained when the adjoint functions are integrated over small timesteps. The adjoint solution is generated using restart files, so it is best to request frequent restarts.

In principle, the obvious timesteps to use are those used in the forward simulation. Unfortunately, the variation of timesteps between optimizer iterations makes this unsuitable unless the parameters are defined on a separate time axis. Thus, for example, the adjoint functions could be integrated at the timestep level, whilst the parameters are defined at some coarser scale. The presence of the constraint functions precludes this as a practical technique; since with fewer parameters than constraint functions, the adjoint system is under determined, making gradient calculations impossible.

Parameter bounds

It is quite important to set realistic parameter bounds for the optimization, using the parameter definition under the [OPTPARS](#) keyword. This prevents unphysical (and unacceptable) values from occurring, and also prevents the simulator from going into tricky convergence regions.

Termination criteria

It is important to set the optimizer tolerances appropriately using the [OPTTUNE](#) keyword. These have been set with broadly reasonable defaults but, if they are changed, then care should be taken not to set them too tight - for example, by allowing termination only when the objective function is changing by too small a fraction.

Guidelines on using the reservoir optimization feature

This section gives you some guidelines on how to use the reservoir optimization feature.

Note: The starting point for the use of the reservoir optimization feature is that you should have an Eclipse 300 model that runs without any problems.

Screening your model for suitability

The first thing you should do is check your model for its compatibility with the requirements of the Reservoir Optimization feature.

Features that the model must include

It should be fully implicit.

It should have reasonably frequent report step output; monthly report output is a good rule-of-thumb.

The adjoint method in the reservoir optimization feature uses report steps, as opposed to timesteps, to calculate the required gradients, so the gradient calculation will lose accuracy if the report steps are too far apart.

Features that the model should not include

The use of the parallel option ([PARALLEL](#) keyword).

The use of fast restarts ([LOAD](#) keyword).

The use of local grid refinements ([CARFIN](#), [EXTFIN](#) or [RADFIN](#) keywords).

The use of grid coarsening ([COARSEN](#) keyword).

The use of the sector modeling option ([SECTOR](#) keyword).

The use of the thermal option ([THERMAL](#) keyword).

AIM, IMPES or IMPSAT formulations.

Any well controls which will compete against the optimizer, for example [ACTION](#) keywords, [WECON](#).

If necessary (and possible) you need to decide if it is possible for you to modify your model to meet these requirements. If you change your model, make sure that it can still be run successfully in Eclipse 300 and that these changes do not affect the results of the simulation dramatically.

If your model is large, then it may be useful to construct a coarse version so that you can determine if the Reservoir Optimization feature gives useful results, before attempting the use the feature on the detailed model.

Deciding the optimization problem you want to solve

You need to decide what objective you are trying to achieve, for example, maximizing discounted oil production total while minimizing water production total.

You also need to decide what constraints (if any) you need to impose while trying to achieve this objective, for example, the water cut must never rise above 10% in certain wells and groups.

Finally you must decide which well control parameters you want to use to try to achieve your constrained objective.

- Well target production/injection rates are only good for optimization purposes as long as the well stays on target production. Once the well falls off target rate control there will be no response to changing the target. In this situation, WBHP is a good control parameter as there will always be a response while the well is open.
- For smart well optimization, the use of valve parameters controlling the cross-sectional area of a valve segment ([WSEGV](#)) is encouraged.
- Typically the calculation of adjoint gradients (sensitivities) will take a time which is less than, but of the order of, the time of the normal simulation.
- Each simulation performed with the control parameters obtained by a model update from the adjoint gradients will take a similar time, and as this is a gradient-based optimization, the use of a line-search algorithm means that there may be several simulations before an optimum is achieved using a given search direction.

The use of constraints and also simple bounds on parameters help to constrain the line-search step size and reduce the number of simulation runs required. Clearly however the overall time to run the optimizer can become a concern. You can control the overall run time as follows:

Now that you have decided your optimization problem, there are a couple of issues for you to consider related to the time it takes for your model to run.

- Control on the maximum number of optimization iterations (gradient evaluations) and simulations is provided using the [OPTDIMS](#) keyword.
- If the optimization problem (that is the objective function, parameters and constraints) is restricted to a particular sector of the model, then you should consider using a model of just that sector to reduce the time required for each simulation.

- If your model has a long history period, and the optimization problem (that is the objective function, parameters and constraints) are only specified in the prediction part of the schedule, then you should consider using a restart file to reduce the time required for each simulation.

Modifying your model for reservoir optimization

To specify your reservoir optimization problem in the model, you need to:

1. Add the `RESOPT` keyword to the `RUNSPEC` section to request the Reservoir Optimization
2. Add the `OPTIMIZE` section keyword before your `END` keyword
3. Add the `OPTFUNC` keyword to the `OPTIMIZE` section to specify your objective function
4. Add the `OPTCONS` keyword to the `OPTIMIZE` section to specify your production/injection constraints.

- a. You should ensure that none of these constraints are violated at the start of the optimization because the optimizer must start from a feasible position honoring all the constraints.

Eclipse 300 will check for this when processing the `OPTCONS` keyword, but it is a good idea to look at the values of constraint quantities in the forward simulation before adding the `OPTCONS` keyword.

For example, if the maximum FWPR in the forward simulation is $2000\text{sm}^3/\text{day}$, then setting an optimization constraint of $1000\text{sm}^3/\text{day}$ is invalid. The starting point for simulation will violate constraints and there is no mechanism to return to a feasible parameter space honoring the constraint.

5. Add the `OPTPARS` keyword to the `OPTIMIZE` section to specify your control parameters.
 - a. You should try to make sure that your initial parameter values lie within the upper and lower limits set by this keyword.

There is an initial feasibility check in Eclipse 300 to ensure that the parameter values in the forward simulation are within these limits, though it is a good idea to look at the range of parameter values in the forward simulation before specifying the limits.

6. Add the `OPTDIMS` keyword to the `OPTIMIZE` section to specify how many optimization iterations and how many simulations are allowed in the Reservoir Optimization feature.

Tip: Initially these limits should both be set to one so that you can perform a simple sensitivity analysis for parameter screening.

7. Add the `RPTOPT` keyword to the `OPTIMIZE` section and request `SEPARATE` output.

This uses more hard-disk space than non-separate output but does allow tracking of the progress of the parameters, constraints and objective function components fully. A separate summary file will be created for each optimization iteration, and these production profiles may be visualized independently in Eclipse Office.

8. As you would in a normal simulation, add summary and reporting output requests for any simulation results you want to track.
 - a. If a production constraint has been added in the `OPTIMIZE` section it is good practice to add the mnemonic for this constraint to the `SUMMARY` section so that you can track whether the constraint is violated in each simulation.

Note: The method used for calculating the adjoint gradients relies on restart files (containing the well and grid block solutions) being generated at each report time during the simulation. Any **RPTRST** keyword mnemonics that turn off the required reporting will be ignored.

Checking your modified model

Now that you have specified the Reservoir Optimization problem in the model, and you have indicated that you just want one gradient calculation performed and no optimization (using the **OPTDIMS** keyword), you should run the model in Eclipse 300 and examine the content of the PRT file.

1. Look for general errors and warnings, in Eclipse 300 these always start with the strings “ @--Error ” and “ @--Warning ”. There should be no errors introduced if the steps above have all been followed.

- This should tell you if any constraints are violated at your starting point. If this is the case, you will get the message:

```
@--Error at 150.0000 Days 4 May 1988
@ Optimization initial point is infeasible
```

The optimizer must stay within the feasible parameter space at all times (except when backtracking from a constraint). If you are violating any constraints, then you either have to change your starting point to make it feasible or you have to change your constraint.

- This will also tell you if your initial parameter values are outside their lower and upper bounds. If this is the case, you will get a message similar to:

```
@--Error at 150.0000 Days 4 May 1988
@ Optimization Parameter 5 is out of range
@ Parameter Name : WBHP
@ Parameter Domain: P1
@ Restart Index : 5
@ Current parameter value is 250.000
@ Lower limit =0.100000E+03, Upper Limit =0.200000E+03
@ Change parameter range using the OPTPARS keyword
```

If you have initial parameter values that are outside their lower and upper bounds, then you must either change your initial parameter values, or you must change your parameter bounds.

2. Look at the sensitivity of your objective to the parameter values. This is indicated by the magnitude of the search direction reported for the parameters. These are reported in a table immediately after the text:

```
=Optimization Parameter Update=====
```

You need to decide whether there is a reasonable response of the objective function to the parameters you specified. (The definition of a reasonable response is largely problem dependent, the expected parameter changes should not all be too close to zero.) If there is not a reasonable response, then you should reconsider your parameter choice as the optimizer will not be able to make much progress with ineffective parameters.

If you need to change the parameters or constraints for any of the reasons given in this section, then you will need to run the Reservoir Optimization again to check the new optimization problem you have specified.

Tracking the progress and results of the optimizer

If you have a feasible starting point, and you are happy with your choice of parameters, then you should change the limits on the number of iterations and simulations given in the `OPTDIMS` keyword. The new values you choose will be largely dependent on the time your model takes to simulate.

Tip: A reasonable choice for the first optimization attempt is a limit of ten on each number.

Then you run the simulation model with the Reservoir Optimization feature. Essentially, this performs a sequence of simulations for you, calculating new gradient directions and performing line-searches when deemed necessary by the optimizer.

You can check the progress of the optimizer in the PRT file. At the end of each simulation, the following information in output:

```
=====
Optimization Iteration: 1 Simulation No: 1
Current Objective Function value: 2702118.0
Prev Valid Objective Function value: 1986265.8
Best Valid Objective Function value: 2702118.0 (at Itn: 1 Sim: 1)
Tolerance in Objective Function : 0.1000E-05
=====
```

This output tells you how many simulations and gradient calculations have been performed so far. It also tells you the current and previous objective function values, the best valid objective function (that is, the largest value for which no constraints have been violated) and the iteration and simulation at which it occurred.

During line-searches, it is possible for the optimizer to step too far in the search direction and hence violate a constraint. If this happens, you will see text such as:

```
Constraints have been overshot using Alpha = 512.00000
Backtracking Alpha to honor constraints.
Proceeding with new Line Search Alpha value = 259.17128
```

The status of any active constraints are reported in a table under the text:

```
|==Optimization: Constraints=====|
```

This table lists the constraints that we are sufficiently close to so that we can consider we are actually on the constraint (that is, have an activity ratio with an absolute value of less than unity).

Active Constraint	Report Index	Current Value	Limiting Value	Activity Ratio
FWPRFIELD	14	0.1443E+03	0.1500E+03	-0.763
FWPRFIELD	15	0.1501E+03	0.1500E+03	+0.019
FWPRFIELD	16	0.1555E+03	0.1500E+03	+0.737

At each simulation, the parameter values (and the update in these values from the previous simulation) are output in a table after the text:

```
=Optimization Parameter Update=====
```

At the end of the optimization, the parameter values that were used in the simulation that gave the best objective function and did not violate the constraints are given in a table after the text:

```
=Optimization Parameters For Best Valid Iteration=====
```

The reason that the optimizer terminated is reported at the end of the PRT file:

```
=====
Optimization Terminating:
```

This will be one of the following:

1. Objective function has converged

This means that the relative change in the objective function over the last two simulations is less than the tolerance specified in the [OPTTUNE](#) keyword

2. Objective function has zero gradient

This means that the average gradient of the objective function with respect to the control parameters is less than the tolerance specified in the [OPTTUNE](#) keyword

3. Control parameters have converged

This means that the average relative parameter change over the control parameters is less than the tolerance specified in the [OPTTUNE](#) keyword

4. No control parameters are active

5. Maximum number of optimization iterations reached.

This means that we have reached the limit on the number of optimization iterations specified in the [OPTDIMS](#) keyword

6. Maximum number of optimization simulation reached

This means that we have reached the limit on the number of optimization iterations specified in the [OPTDIMS](#) keyword

If you ran the optimizer using the [SEPARATE](#) mnemonic in the [RPTOPT](#) keyword, then the Reservoir Optimization produces separate results files for each iteration in the optimization. These results files have names of the form `BASE_01.xxx`, `BASE_02.xxx` and so on where `BASE.DATA` is the base data-set, and `_01`, `_02` and so on label the optimization iteration. You can track the progress of the objective function, constraints and parameters at each optimization iteration by plotting their equivalent summary vectors (if you have requested these) in Results Viewer in Eclipse Office. If you intend to view the results in this way, then you should also specify the [CLRFILES](#) mnemonic in the [RPTOPT](#) keyword so that the results files from any previous optimizations are removed before starting this optimization, otherwise the presence of these old results files can cause confusion.

[De Montleau et al \(2006\)](#) gives some examples of reservoir optimization studies.

Summary of keywords (Eclipse 300)

RUNSPEC

- [RESOPT](#) Activate the Reservoir Optimization License.

SCHEDULE section

- [RPTRST](#) Set the restart output frequency using [BASIC](#) and [FREQ](#)

OPTIMIZE section

Keyword	Description
OPTCONS	Specify the optimization constraint functions
OPTDIMS	Specify the key optimization dimensions
OPTFUNC	Specify the optimization objective function
OPTLOAD	Hot-start a reservoir optimization simulation
OPTOPTS	Specify the optimization options (for example which optimizer to use)
OPTPARS	Specify the optimization control parameters
OPTTUNE	Define tuning for the optimization (for example, tolerances)
RPTOPT	Request the style of optimization output

Miscellaneous

- [Item 77](#) of the [OPTIONS3](#) keyword - compute and output constraint function gradients

Example

The following is the OPTIMIZE section of a reservoir optimization example data set.

The Reservoir Optimization license has been activated by the presence of the [RESOPT](#) keyword in the RUNSPEC section. Restart files have been requested every calendar month.

```

-----
OPTIMIZE
-----

OPTDIMS
-- Maximum no.   Maximum no.
-- Iterations    Simulations
   30             100 /

OPTTUNE
-- Constraint Tolerance
   0.05 /

OPTFUNC
-- O.F.          O.F.
-- Component     Domain
   'FOPT'        'FIELD' /
/

OPTPARS
-- Parameter     Parameter   Lower   Upper
-- Mnemonic      Domain      Limit   Limit
   'WBHP'        'P1'        100.0   350.0 /
   'WBHP'        'P2'        100.0   350.0 /
/

OPTCONS
-- Constraint    Constraint   Constraint   Constraining
-- Mnemonic      Domain        Type         Value
   'FOPR'        'FIELD'       '<'          3015 /
   'FGPR'        'FIELD'       '<'          665000 /
/

```

In this example:

- The objective is to maximize the field oil production total (specified using the [OPTFUNC](#) keyword).
- This is subject to limits of 3015 SM³/DAY on the field oil production rate and 665000 SM³/DAY on the field gas production rate (specified using the [OPTCONS](#) keyword).
- These constraints are considered to be met if the simulated value lies within 1 part in 20 of the limiting value (specified using the [OPTTUNE](#) keyword).
- The optimizer tries to achieve this objective by varying the bottom hole pressure in wells P1 and P2 between 100 and 300 BARSA at each reporting interval over the whole time period of the simulation (specified using the [OPTPARS](#) keyword).
- The optimizer is allowed a maximum of 30 gradient simulations or 100 forward simulations (whichever happens first) to try to achieve this objective (specified using the [OPTDIMS](#) keyword).

Gradient option

x	Eclipse 100
x	Eclipse 300

The gradient option computes gradients of the solution (for example, pressure in a well) with respect to various property parameters that you define, such as horizontal permeability in a layer.

The SimOpt program uses the gradients computed by the simulator to achieve an efficient regression during a history match or sensitivity analysis, varying the property parameters via modifiers.

These property parameter modifiers are usually set using [HMMULTxx](#) on the various gradient regions, set in [HMxxxxxx](#), unless otherwise specified.

- X-direction transmissibility multipliers input through parameter [HMMULTX](#) applied to region [HMTRANX](#)
- Y-direction transmissibility multipliers input through parameter [HMMULTY](#) applied to region [HMTRANX](#)
- Z-direction transmissibility multipliers input through parameter [HMMULTZ](#) applied to region [HMTRANZ](#)
- X and Y-direction transmissibility multipliers input through [HMMLTXY](#) applied to region [HMTRNXY](#)
- Dual porosity sigma factor multipliers input through keyword [HMMULTSG](#) applied to region [HMSIGMA](#).
- Pore volume multipliers input through parameter [HMMULTPV](#) applied to region [HMPORVM](#)
- Analytic aquifer strengths through keywords [HMMLCTAQ](#) applied to [HMAQUCT](#) (Carter Tracy), and [HMMLFTAQ](#) applied to [HMAQUFET](#) (Fetkovich).
- Numerical aquifer pore volume, permeability and transmissibilities input through keyword [HMMLAQUN](#) applied to region [HMAQUNUM](#).
- Transmissibility multipliers across nominated faults through [HMMULTFT](#) applied to [HMFAULTS](#).
- X-direction permeability multipliers input through parameter [HMMLTPX](#), applied to region as in [HMPERMX](#).
- Y-direction permeability multipliers input through parameter [HMMLTPY](#), applied to region as in [HMPERMY](#).
- Z-direction permeability multipliers input through parameter [HMMLTPZ](#), applied to region as in [HMPERMZ](#).
- X and Y-direction permeability multipliers input through [HMMLTPXY](#), applied to region as in [HMPRMXY](#).
- Saturation table end-points: [HMSWCR](#), [HMSGCR](#), [HMSOWCR](#), [HMSOGCR](#), [HMSWL](#).
- Multipliers on vertical scaling of relative permeability: [HMKRO](#), [HMKRW](#), [HMKRG](#), [HMKRORW](#), [HMKRORG](#).
- Multipliers on vertical scaling of capillary pressure: [HMPCW](#), [HMPCG](#).
- Rock compressibility multipliers input through [HMMROCK](#) applied to [HMROCK](#), and modifiers on rock compaction parameters through [HMMROCKT](#) applied to [HMROCKT](#).
- Inter-region transmissibility multipliers between nominated regions using keyword [HMMMREGT](#) applied to [HMMULRGT](#).
- Multipliers on the well productivity index using keyword [HMWPIMLT](#).

- Multipliers on the well connection transmissibility factor and skin using the keyword [HMMLTWCN](#) applied to the connections specified in [HMWELCON](#).

All the above parameters may be used with Eclipse 100. In this release of Eclipse 300 pore volume, transmissibility and permeability parameters are available.

The derivatives are written to a file (extension .HMD) which can be interpreted by SimOpt.

The data contained within the .HMD file is generally encoded and is meaningless unless the SimOpt program is used. A facility for unencoded gradients is also available and is controlled by the licensing system. Please contact your Eclipse account manager for details.

Output from the gradient option

Gradients of observable well quantities (for example, WBP, WOPR) are written to the HMD file at each timestep and these can be used by SimOpt to match against observed data using a gradient based optimization approach.

The [GRADWELL](#) keyword specifies a list of well summary vectors for which derivatives are required.

Wells requiring gradient output are nominated in the [RPTHMW](#) keyword

Eclipse 100

It is also possible to output gradients of simulated RFT data to the GRFT file to match against observed RFT tests by using the [GRADRFT](#) keyword.

Eclipse 100

If required, it is also possible to output gradients of reservoir grid block solutions (SWAT, SGAS, SOIL, RS, RV, PRESSURE, DENO, DENW, DENG, PORO), as well as seismic impedances (ACIP, ACIS), to the RESTART files, using the [GRADRESV](#) keyword. Currently, this is only possible using the licensing system. Please contact your Eclipse account manager for details. The arguments available in the GRADRESV keyword are:

Argument	Description
SWAT	Water saturation
SGAS	Gas saturation
SOIL	Oil saturation
RS	Dissolved gas-oil ratio
RV	Vaporized oil-gas ratio
PRESSURE	Oil phase pressure
DENO	Oil phase gravity density at reservoir conditions
DENG	Gas phase gravity density at reservoir conditions
DENW	Water phase gravity density at reservoir conditions
PORO	Porosity at reservoir conditions
ACIP	Pressure-wave impedance
ACIS	Shear-wave impedance

Using the facility

The facility is initiated by setting the appropriate number of property parameters for which gradients are calculated in the RUNSPEC keyword [HMDIMS](#).

You should supply region tables in the `REGIONS` section using keywords `HMPORVM`, `HMTRANX` etc. to delimit the property parameter regions on the simulation grid.

Eclipse 100 Alternatively, for faults which have been specified using the `FAULTS` keyword, property parameters may be requested using the `HMFAULTS` keywords in the `GRID` section.

Eclipse 100 Gradients for analytic aquifer properties may be requested by using the keywords `HMAQUFET` or `HMAQUCT` in the `SOLUTION` section, whereas gradients for numerical aquifer properties may be requested by using the keyword `HMAQUNUM` in the `GRID` section.

Eclipse 100 Gradients for rock compaction and compressibility parameters can be obtained by specifying `HMROCK` and `HMROCKT` in the `PROPS` section.

Eclipse 100 To obtain gradients of the solution with respect to inter-region transmissibility multipliers declared using keyword `MULTREGT` in the `GRID` section, the `HMMULRGT` keyword can be used in the `GRID` section.

Eclipse 100 To obtain gradients of the solution with respect to well connection transmissibility factor and skin multipliers declared using keyword `HMMLTWCN` in the `SOLUTION` section, the `HMWELCON` keyword can be used in the `SOLUTION` section.

For each gradient property parameter, Eclipse calculates derivatives of well-flow quantities (as specified with the `GRADWELL` keyword) for all active wells. This information is written to the file `ROOT.HMD`, where `ROOT` is the base name of the run.

Eclipse 100 Additional information can be written out to the `ROOT.HMD` file if requested; for example, refer to the `GRADRESV` keyword.

Eclipse 100 If a satisfactory match has been obtained using modifiers based on the gradient analysis, then the model may be used for prediction without the overhead in calculating the derivatives, simply by specifying the `NOHMD` (or `NOHMO`) keyword in the `SCHEDULE` section (which switches off all gradient calculations). This is the mechanism for a 'evaluation' run in SimOpt.

Current restrictions

Eclipse 100 • The gradient option may not at present be used with the parallel option or with reservoir coupling.

Eclipse 100 • The effects of active tracers (API, brine and the temperature option) and environmental tracers or the foam model are not currently accounted for in a rigorous fashion while calculating gradients. Changes in solution properties caused by active tracers first influence the gradient calculation one timestep after they influence the fluid flow properties, that is the active tracers are treated explicitly in the gradient calculation. In most cases, this effect should be very small and can often be neglected so that it is still possible to perform a sensitivity run to screen redundant parameters or enlist a regression run. Eclipse no longer disallows a run with active tracers, but the gradients will not be fully accurate.

Eclipse 100 • LGRs employing local timestepping are not currently available in gradient runs. However, the gradient option is fully compatible with the LGR-in-place option, and also with the grid coarsening option. In many applications, it is possible to get an initial sensitivity analysis on a coarsened grid to screen for redundant parameters.

Eclipse 100 • Gradients cannot at present be obtained with the vertical equilibrium option (`VE` or `COMPVE`), with the gravity drainage option (`GRAVDR` or `GRAVDRM`) in dual porosity runs, or with the water-induced rock compaction options. This is because the gradient calculation relies on the fully implicit nature of Eclipse; the `VE` option treats the pseudo-capillary pressure explicitly and the extra derivative terms are not accounted for. Eclipse no longer disallows a run with `VE` or gravity drainage, but the gradients will not be fully accurate as approximations have been made in the gradient calculation.

Eclipse 300

- Gradients cannot at present be obtained for multisegment injector wells.
- Gradients cannot be used with the grid coarsening option.

Current Uses (Eclipse 300)

- It is possible to use the gradient option in both Black Oil and Compositional runs.
- In addition, this option can be used in Fully Implicit and AIM (Adaptive IMplicit) modes.
- The gradient option may be used with the parallel option in Eclipse 300.

Example

Example 1

To set three transmissibility multiplier parameters (with $NX=3$ $NY=3$ $NZ=3$), set the number of parameters, gradient regions and subregions using RUNSPEC keyword `HMDIMS`

```
HMDIMS
 1 1 3 /
```

and set appropriate keywords in the REGIONS section

```
HMTRANZ
3*0 3*1 3*0 /
COPY
'HMTRANZ' 'HMTRANX' /
/
EQUALS
'HMTRANY' 9*1 /
/
```

Example 2

To set one pore volume multiplier parameter (with $NX=3$ $NY=3$ $NZ=3$), set the number of parameters, gradient regions and subregions using RUNSPEC keyword `HMDIMS`

```
HMDIMS
 1 1 1 /
```

and set appropriate keyword in the REGIONS section

```
HMPORVM
3*0 3*1 3*0 /
```

Data file overview

RUNSPEC section overview

The following keywords are specific to the gradient option

Keyword	Description
FMTHMD	Declares that the HMD output file containing the derivatives is to be formatted. The default is unformatted.
HMDIMS	Specify the dimensions for the gradient option.
RPTHMD	Controls on output to the HMD file.
UNCODHMD	Requests that the HMD file contains unencoded gradients.

Example 1 (Eclipse 100)

In the following example there is a maximum of 3 gradient regions, each with up to 4 subregions with only 5 gradients required and 5 faults (with gradients required for two faults). There are two analytic aquifers with up to 100 grid block connections each and a default influence table.

```

FAULTDIM
--MFSEGS
  5 /
AQUDIMS
--MXNAQN  MXNAQC  NIFTBL  NRIFTB  NANAQU  NCAMAX
  0         0       1       36       2       100 /
HMDIMS
--MHISTM  NHMSTR  NHRPAR  MFHIST  MHAQUF
  3        4       5       2       2 /
RPTHMD
  0 1 1 1 /
UNCODHMD

```

Example 2

In the following example there is a maximum of 3 gradient regions, each with up to 4 subregions with only 5 gradients required.

```

HMDIMS
--MHISTM  NHMSTR  NHRPAR
  3        4       5 / UNCODHMD

```

Summary section overview (Eclipse 100)

Gradient calculation performance keywords

The following keywords measure performance of the gradient calculation and take no data:

Keyword	Description
TCPUTSHT	CPU time per timestep for all the gradient calculations.
TCPUHT	Total CPU time for all the gradient calculations.

The following keywords measure performance for individual gradient parameters:

Keyword	Description
TCPUTSH	CPU time per timestep for each gradient calculations.
TCPUH	Total CPU time for each gradient calculations.
HLINEARS	Number of linear iterations for each gradient calculation
HSUMLINS	Cumulative number of linear iterations for each gradient parameter.

The keywords TCPUTSH, TCPUH, HLINEARS and HMSUMLINS should each be followed by a list of gradient parameter names, terminated by a slash (/). If the entire list is defaulted, the relevant quantities are output for all parameters.

Allowable parameter names are:

- Any gradient region keywords `HMxxxxxx` entered in the REGIONS section
- `HMFAULTS`
- `HMAQUCT`, `HMAQUFET` and `HMAQUNUM`
- `HMROCK` etc.
- `HMMULRGT`

Example

Total CPU time spent on the gradient calculations

```
TCPUH
/
```

Number of linear iterations to solve for each parameter associated with HMPORVM and each parameter for Carter-Tracy aquifers declared using HMAQUCT.

```
HLINEARS
'HMPORVM' 'HMAQUCT' /
```

```
HMSUMLINS
'HMPORVM' 'HMAQUCT' /
```

Special extensions

When unencoded derivatives are enabled, some extra summary keywords are available: these mnemonics have first character 'Q' and are only activated if the HMD file is unencrypted. The Q* mnemonics should be specified at the end of the summary section, after all the normal Eclipse SUMMARY keywords.

Analytic and Numerical Aquifer derivatives

These are analogous to AAQR, AAQT and AAQP for analytic aquifers (and ANQR, ANQT, ANQP for numerical aquifers):

Derivative of influx rate for analytic aquifers:

```
QAAQR
iaq ihm /
```

Derivative of total influx for analytic aquifers:

```
QAAQT
iaq ihm /
```

Derivative of aquifer pressure (Fetkovich only):

```
QAAQP
iaq ihm /
```

Derivative of influx rate for numerical aquifers:

```
QANQR
iaq ihm /
```

Derivative of total influx for numerical aquifers:

```
QANQT
iaq ihm /
```

Derivative of average numerical aquifer pressure:

```
QANQP
iaq ihm /
```

where

iaq = aquifer number as specified in AQUCT, AQUFETP or AQUNUM (this may be defaulted to refer to all aquifers,

ihm = number of derivative for aquifer (this item may be defaulted to refer to all parameters for a given aquifer).

For a Carter-Tracy aquifer, *ihm* is used as follows:

1. derivative with respect to permeability multiplier
2. derivative with respect to opening angle multiplier
3. derivative with respect to aquifer depth.

For a Fetkovich aquifer:

1. derivative with respect to multiplier on initial water volume
2. derivative with respect to Productivity Index multiplier
3. derivative with respect to aquifer depth.

For a Numerical aquifer:

1. derivative with respect to multiplier on total aquifer pore volume
2. derivative with respect to aquifer permeability multiplier
3. derivative with respect to aquifer to grid transmissibility multiplier.

The derivatives are only calculated when the aquifer parameters HMAQUCT, HMAQUFET or HMAQUNUM are selected.

Example

```
QAAQR
1 2 /
2 2 /
3 1 /
/
```

Derivatives of well quantities

For derivatives of WBP with respect to all gradient parameters

```
QWBP
WELL1 /
WELL2 /
/
```

The available arguments are:

Keyword	Description
QWBP	derivative of WBP
QWBP4	derivative of WBP5
QWBP9	derivative of WBP9
QWOFR	derivative of WOPR or WOIR
QWWFR	derivative of WWPR or WWIR
QWGFR	derivative of WGPR or WGIR
QWWCT	derivative of WWCT
QWGOR	derivative of WGOR
QWBHP	derivative of WBHP

The corresponding quantities in local grids are preceded by the letter L, for example:

```
LQWBP
LGR1 WELL1 /
LGR2 WELL2 /
/
```

To get derivatives of WBP for all parameters at each well:

```
QWBP
/
WBP
/
RUNSUM
SEPARATE
RPTSMRY
1 /
```

Derivatives of grid block quantities

The available mnemonics are:

Argument	Description
QBPR	derivative of BPR
QBR5	derivative of BRS
QBRV	derivative of BRV

Argument	Description
QBOSAT	derivative of BOSAT
QBWSAT	derivative of BWSAT
QBGSAT	derivative of BGSAT
QBNSAT	derivative of BNSAT

For example:

```
QBPR
  7 1 1 idm /
 17 1 1 idm /
/
```

where *idm* is the index of the derivative. If *idm* is defaulted, then block derivative for all parameters are obtained. The *idm* parameter is the internal Eclipse numbering of each parameter. Usually, *idm* is defaulted and gradients for every parameter are output.

The corresponding quantities in local grids are preceded by the letter L, for example:

```
LQBPR
  'LGR1' 7 1 1 idm /
  'LGR2' 17 1 1 idm /
/
```

Output from the Gradient option

The derivatives of wellbore quantities generated by the Gradient option are output to the file `ROOTNAME.HMD` (see ["File handling in Eclipse"](#)). The output is generated by the standard Eclipse output routines and typically looks like:

```
'CODESTAT'      1 'CHAR'
'NOTCODED'
'E100VER'       1 'INTE'
  901
'E100INFO'      4 'CHAR'
'27/05/96' '12:37:00' 'test' 'test'
'E100ORIG'      3 'CHAR'
'May 26' '1996' '03:06:53'
'GRADWELL'      9 'CHAR'
'WOPR' 'WWPR' 'WGPR' 'WWCT' 'WGOR' 'WBP' 'WBP4'
'WBP5' 'WBP9'
'TIMESTEP'      2 'DOUB'
  0.18260000000000D+04 0.31000000000000D+02
'NAMES'        5 'CHAR'
'PRO' 'HMPORVM' 'PRODUCER' 'OPEN'
'HMHEAD1'      2 'INTE'
  1 6
'DERIVS'       9 'DOUB'
  0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
  0.00000000000000D+00 0.00000000000000D+00 0.26393554627632D+00
  0.26588984672034D+00 0.26491269649833D+00 0.26491269649833D+00
'VALUES'       9 'DOUB'
  0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
  0.00000000000000D+00 0.00000000000000D+00 0.20023871765242D+03
  0.20038969805460D+03 0.20031420785351D+03 0.20031420785351D+03
'NAMES'        5 'CHAR'
'PRO' 'HMTRANX' 'PRODUCER' 'OPEN'
'HMHEAD1'      2 'INTE'
  1 1
'DERIVS'       9 'DOUB'
  0.00000000000000D+00 0.00000000000000D+00 0.00000000000000D+00
  0.00000000000000D+00 0.00000000000000D+00 -
```

```

0.15196542044025D+00
-0.18932798370102D+00  -0.17064670207064D+00  -0.17064670207064D+00
'NAMES      '      5  'CHAR'
'PRO        ' 'HMTRANZ' '      ' 'PRODUCER' 'OPEN      '
'HMHEAD1    '      2  'INTE'
'            '      4
'DERIVS     '      9  'DOUB'
0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
0.00000000000000D+00  0.00000000000000D+00  0.35438853150093D-02
0.19550833005848D-02  0.27494843077971D-02  0.27494843077971D-02

```

Record 'TIMESTEP'

This record consists of 2 items (DOUBLE PRECISION)

1. Current time
2. Last timestep used

Subsequent records are written in the order shown in the following figure:

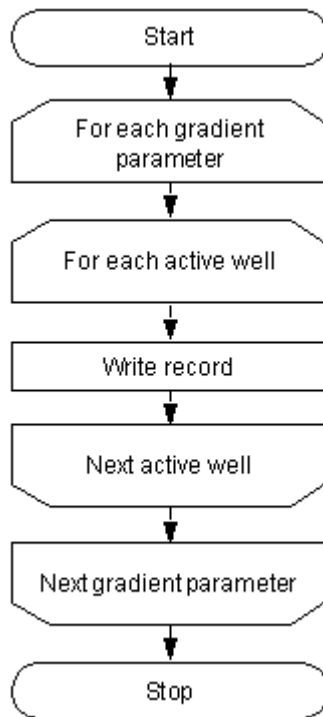


Figure 13.4. Order for writing records

Record 'NAMES'

This record consists of 5 items (CHARACTER*8)

1. Well name
2. Name of Gradient parameter. This is the keyword name associated with the parameter. For example:

HMxxxxxx in the REGIONS section (for example, HMPORVM, HMPERMX etc.

HMFAULTS in the GRID section

- HMMULRGT in the GRID section
- HMAQUFET in the SOLUTION section
- HMAQUCT in the SOLUTION section
- HMKRW etc. in the PROPS section
- HMSWCR etc. in the PROPS section
- HMROCK etc. in the PROPS section
- 3. Local grid name or blank for a well in the global grid
- 4. Status of the well:
 - 'PRODUCER' or 'INJECTOR'
- 5. Status of the well: 'OPEN' or 'STOPPED' or 'SHUT'

Note: Item 2: Only REGIONS section keyword HMPORVM is available at present.

Record 'HMHEAD'

This record consists of three items (INTEGER)

1. For HMxxxxxx(REGIONS parameters) - the index to the current subregion
 - For HMFAULTS - the number of the fault (in the data order)
 - For HMAQUFET - the number of the Fetkovich aquifer
 - For HMAQUCT - the number of the Carter-Tracy aquifer
 - For HMAQUNUM - the number of the numerical aquifer
 - For HMROCK - the number of the PVTNUM or SATNUM region
 - For HMROCKT - the number of the ROCKNUM region
 - For HMMULRGT - the number of the first MULTNUM region
2. The type of Gradient parameter:

1	HMTRANX
2	HMTRNXY
32	HMTRANY
4	HMTRANZ
5	HMSIGMA
6	HMPORVM
7	HMTRNX-
8	HMTRNY-

Eclipse 300

9	HMTRNZ-	
12	HMPERMX	
13	HMPRMXY	
14	HMPERMY	
15	HMPERMZ	
101	HMFAULTS	
102	HMAQUFET	initial water volume multiplier
103	HMAQUFET	productivity index multiplier
104	HMAQUFET	aquifer depth
105	HMAQUCT	permeability multiplier
106	HMAQUCT	opening angle multiplier
107	HMAQUCT	aquifer depth
108	HMWPIMLT	
110	HMMULRGT	
131	HMAQUNUM	pore volume multiplier
132	HMAQUNUM	permeability multiplier
133	HMAQUNUM	transmissibility multiplier
297	HMSWCR	
298	HMSGCR	
299	HMSOWCR	
300	HMSOGCR	
301	HMSWL	
306	HMKRG	
307	HMKRO	
309	HMKRWR	
310	HMKRGR	
311	HMKRORG	
312	HMKRORW	

- 313 HMSWLPC
- 314 HMSGLPC
- 315 HMPCW
- 316 HMPCG
- 1041 HMROCK
- 1042 HMROCKT

3. For HMMULRGT - the number of the second MULTNUM region
Not used for all other parameters.

Eclipse 300

Note: Item 1: Only REGIONS section keyword HMPORVM is available at present

Record 'DERIVS'

This record consists of a variable number of items (DOUBLE PRECISION). The record contains the derivatives with respect to each Gradient parameter of nominated well quantities. By default these are:

- Eclipse 300
1. Oil rate at surface conditions
UNITS: sm³/day (METRIC), stb/day (FIELD), scc/hr (LAB)
- Eclipse 300
2. Water rate at surface conditions
UNITS: sm³/day (METRIC), stb/day (FIELD), scc/hr (LAB)
- Eclipse 300
3. Gas rate at surface conditions
UNITS: sm³/day (METRIC), Mscf/day (FIELD), scc/hr (LAB)
- Eclipse 300
4. Water cut
UNITS: sm³/sm³ (METRIC), stb/stb (FIELD), scc/scc (LAB)
- Eclipse 300
5. Gas Oil ratio
UNITS: sm³/sm³ (METRIC), Mscf/stb (FIELD), scc/scc (LAB)
6. 1 point pressure average (see WPAVE and WWPAVE)
UNITS: barsa (METRIC), psia (FIELD), atma (LAB)
7. 4 point pressure average (see WPAVE and WWPAVE)
UNITS: barsa (METRIC), psia (FIELD), atma (LAB)
8. 5 point pressure average (see WPAVE and WWPAVE)
UNITS: barsa (METRIC), psia (FIELD), atma (LAB)
9. 9 point pressure average (see WPAVE and WWPAVE)
UNITS: barsa (METRIC), psia (FIELD), atma (LAB)

This can be changed: see the comments below on [GRADWELL](#).

Record 'VALUES'

This record consists of a variable number of items (DOUBLE PRECISION).

This record contains the values of the well quantities referred to in the 'DERIVS' record. This is only output once for each well at each step, so if there are several Gradient parameters, the values are only output for the first parameter. The reported rates are production rates, so a negative sign indicates injection of the pertinent phase. When a well is shut, the 'DERIVS' and 'VALUES ' records are not calculated and may be reported as zeroes.

Eclipse 100

As writing all this information can be quite voluminous, the amount of output may be configured using the [RPTHMD](#) keyword (to supply output only for working producers, etc.) or by the [RPTHMW](#) keyword (to switch off output for nominated wells), or only at report times, etc.

This data is used by SimOpt. Generally, the derivatives are unformatted and encoded (though the 'VALUES' record is not encoded). The 'CODESTAT' record outputs whether the file is encoded or not. SimOpt must be used to interpret the derivatives. However, the output of unencoded derivative information is controlled by the password mechanism for Eclipse software - please contact your Eclipse account manager for more information.

The following records provide version control:

- Eclipse 100
The records E100VER, E100INFO and E100ORIG provide version control. E100VER is the version of Eclipse; E100INFO lists the date and time at which the HMD file was created together with the data set root name; and E100ORIG reports the date and time at which the current copy of Eclipse was built.
- Eclipse 300
The records E300VER, E300INFO and E300ORIG provide version control. E300VER is the version of Eclipse; E300INFO lists the date and time at which the HMD file was created together with the data set root name; and E300ORIG reports the date and time at which the current copy of Eclipse was built.
- The convert utility may be used to convert from unformatted to formatted if required; the formatted HMD file has the extension FHMD (UNIX) or FHM (PC). The RUNSPEC keyword [FMTHMD](#) requests a formatted Derivatives file.

Alternative output of well quantities

The above default quantities loaded into the 'DERIVS' and 'VALUES' records may be inappropriate. The [GRADWELL](#) keyword allows you to specify a list of mnemonics (such as WBP, WWCT and WBHP) to be output for each well. If the GRADWELL keyword has been used, then extra items are written at the top of the file to specify the GRADWELL data.

For example,

```
'GRADWELL'      3 'CHAR'  
'WBP'           'WCT'           'WBHP'
```

Eclipse 100

By default, if GRADWELL is not specified, then nine items are written as in the above example.

When a well connects across poorly-communicating layers which are not in vertical equilibrium, the WBP mnemonics are appropriate. It will be necessary to output completion properties to match measured RFT data.

Receiving HMD files from Eclipse through OpenEclipse (Eclipse 100)

The HMD file data are sent back from Eclipse to the master process if IHEAD(14) = 1 or 2 in the run header data, when using the OpenEclipse protocol.

Eclipse sends the HMD file in a set of message packets:

1. A 'MESS' header with keyword 'GRAD'
2. Derivatives for each parameter
3. A 'MESS' header with keyword 'END'

The above packets are within the loop over all parameters at each output time. An optional 'MESS' header signifies the HMD output has finished before run termination.

The RFT file and RFT gradients are sent back from Eclipse to the master process if IHEAD(16) = 1 or 2 in the run header data, when using the OpenEclipse protocol.

Eclipse sends the RFT data in a set of message packets:

1. A 'MESS' header with keyword 'RFT'
2. RFT data for a particular well
3. A 'MESS' header with keyword 'END'

The RFT gradients are similar:

1. A 'MESS' header with keyword 'GRADRFT'
2. RFT gradient data for a particular well and parameter
3. A 'MESS' header with keyword 'END'.

The above packets are within the loop over all parameters at each output time. An optional 'MESS' header 'GFTEND' signifies the RFT gradients output has finished before run termination.

Reservoir gradients (Eclipse 100)

It is also possible to request output of gradients of reservoir pressures, saturations, GOR/OGR and densities for example. These may be used to form an objective function in conjunction with a Petro-elastic Model in order to compare predicted seismic response. Such a case might be the calculation of seismic impedance.

Furthermore, it is now possible to request output of gradients of pressure-wave impedance and shear-wave impedance, if the appropriate keywords have been specified in the data set to utilize the Petro-elastic Model inside the simulator. The information output is controlled by the password mechanism for Eclipse software - please contact your Eclipse account manager for more information.

Output of derivatives of grid block pressures and saturations for example, is controlled by the GRADRESV keyword.

A record 'GRADRESV' is written to the HMD and RESTART files to list the mnemonics. For example, referring to the above example

```
'GRADRESV'      2 'CHAR'
'GPRES' 'GSWAT' '/'
```

or

```
'GRADRESV'      4 'CHAR'
'GPRES'      'GSOIL'      'GSWAT'      'GSGAS'      /
/
```

The solution gradient data is written to the `RESTART` file in the following manner (see *Eclipse File Formats Reference Manual* for details of how restart data is written from Eclipse).

Header `GRADSOL` indicates a solution gradient mnemonic:

For example

```
'GRADSOL'      6 'CHAR'
'HMTRANZ'      'GRADZONE'      '1'      '1'      '3'      /
/
```

The record `GRADSOL` has up to 10 items:

1. Parameter name
2. Keyword name associated with parameter

' GRADZONE '	for regional parameters defining gradzones
' FAULTS '	for faults
' AQUCTS '/' AQUFET '	for aquifers
' SATNUM '	relative permeability shape parameters
' PVTNUM '/' SATNUM '/' ROCKNUM '	for rock parameters
' MULTNUM '	for inter-region transmissibility multipliers
3. Feature name: Name of fault for fault parameters

Blank	for regional parameters
Blank	for all other parameters
4. Parameter type. This is described under `HMHEAD1`.
5. Region ID or Feature number. This is the subregion number for regional parameters.

Fault number	for fault parameters
Aquifer number	for aquifer parameters
Saturation table number	for relative permeability shape parameters
Rock table number	for rock parameters
First region number	for inter-region connection parameters
6. Associated phase. This is a pointer internal to Eclipse.
7. Alternate region ID. This is the second region number for inter-region connection parameters

Blank	for all other parameters
-------	--------------------------

Items 8 to 10 Not used at present.

Following each ' GRADSOL ' record, the solution gradients are written in the following form:

```

'GRADSOL ' 6 'CHAR'
'HMTRNXY ' 'GRADZONE' ' '3' '3' '1'
'GPRES_13' 300 'REAL'
0.45278375E+03 0.40524387E+03 0.35893173E+03 -0.87420359E+01
-0.12985765E+02 -0.22687635E+02 -0.40819241E+02 -0.70829422E+02
-0.11564088E+03 -0.16381274E+03 0.19503807E+03 0.19354639E+03
0.18682153E+03 -0.16461343E+01 -0.25963242E+01 -0.47793212E+01

```

The 'GPRES_nn' mnemonic indicates oil pressures. The names in GRADRESV have been preceded by 'G' so they are distinct from names 'SGAS', 'SWAT' that Eclipse use for the solution values.

The header is of the form 'G//name(1:4)//number: Characters 5 to 8 in the header is just a unique number such that each GPRES *** mnemonic is unique to any post-processing application that does not require the 'GRADRESV' data.

Gradient option formulation

A brief description of the method for deriving gradients in the simulator is outlined in this section. For this overview, the focus is on the fully implicit method.

Obtaining gradients in Eclipse

At each simulator timestep, Eclipse forces the fully implicit residual in each simulator grid block to be zero (see "[Formulation of the equations](#)").

This can be written as

$$R(X_{t+dt}, X_t, \lambda) = \frac{M_{t+dt}(X_{t+dt}, \lambda) - M_t(X_t, \lambda)}{dt} + F_{t+dt}(X_{t+dt}, \lambda) + Q_{t+dt}(X_{t+dt}, \lambda) = 0 \quad \text{Eq. 13.24}$$

where

the subscripts t and $t + dt$ denote quantities at the current and advanced time, with a simulator timestep of dt ,

M is the fluid-in-place in each simulator grid block,

F is the net flow rate into neighboring grid blocks,

Q is the net flow rate into wells during the timestep,

λ is any gradient parameter,

X is the set of solution variables in each grid block.

For Eclipse 100, X represents pressure and saturations, and for Eclipse 300, X represents pressure and molar densities.

Differentiating equation 13.24 throughout with respect to λ yields

$$0 = \frac{dR}{d\lambda} = \frac{\partial R}{\partial \lambda} + \frac{\partial R}{\partial X_{t+dt}} \cdot \frac{\partial X_{t+dt}}{\partial \lambda} + \frac{\partial R}{\partial X_t} \cdot \frac{\partial X_t}{\partial \lambda} \quad \text{Eq. 13.25}$$

which can be written as

$$J \cdot \frac{\partial X_{t+dt}}{\partial \lambda} = -\frac{\partial R}{\partial X_t} \cdot \frac{\partial X_t}{\partial \lambda} - \frac{\partial R}{\partial \lambda} \quad \text{Eq. 13.26}$$

where the Jacobian $J = \partial R / \partial X_{t+dt}$ is the same as that used to solve the reservoir equations, so no additional matrix setup is required. Therefore, for each gradient parameter λ selected, a set of linear equations, equation 13.26 must be solved at each simulator timestep to determine the gradient $\frac{\partial X_{t+dt}}{\partial \lambda}$.

When there are several gradient parameters, each requires the solution of the same set of linear equations, equation 13.26 but with different right hand sides. The right hand sides contain two terms. The first is the product of $\partial R / \partial X_t$, which is independent of λ , and the previously calculated gradient at the old timestep, $\partial X_t / \partial \lambda$.

For a fully implicit discretization the second term in the right hand side of equation 13.26 is given by:

$$\frac{\partial R}{\partial \lambda} = \frac{1}{dt} \left(\frac{\partial M_{t+dt}}{\partial \lambda} - \frac{dM_t}{d\lambda} \right) + \frac{\partial F_{t+dt}}{\partial \lambda} + \frac{\partial Q_{t+dt}}{\partial \lambda} \quad \text{Eq. 13.27}$$

Evaluation of equation 13.27 depends on the parameter λ , and additional code is required for each new parameter. If, for example λ is a pore volume multiplier, then it would appear linearly in the accumulation terms, and not appear in the flow or well terms. Therefore equation 13.27 could be evaluated as:

$$\frac{\partial R}{\partial \lambda} = \frac{1}{dt} \left(\frac{\partial M_{t+dt}}{\partial \lambda} - \frac{dM_t}{d\lambda} \right) = \frac{1}{\lambda} \frac{(M_{t+dt} - M_t)}{dt} \quad \text{Eq. 13.28}$$

where the right hand side has already been calculated in evaluating the residual equation 13.24.

Regression in SimOpt using Eclipse gradients

The solution $\partial X / \partial \lambda$ of equation 13.27 consists of derivatives of solution variables in each grid block. These are converted to derivatives of meaningful observable well quantities $\partial W_{calc} / \partial \lambda$ such as well block pressure and well flow rates. SimOpt uses these quantities to calculate derivatives and the Hessian of the objective function with respect to the gradient parameters for use in the regression algorithm.

It is important to note that Eclipse returns gradients $\partial X / \partial \lambda$ around $\lambda = 1$ for multiplier-based parameters (such as HMPORVM, HMPERMV and HMKRW, in HMxxxxxx). During a gradient-based minimization loop, a sequence of multiplicative modifiers $\lambda_i, i = 1, \dots, v$ may be applied to the base data set whose effect is cumulative, so that after v iterations, the cumulative modifier is

$$\Lambda_n = \prod_{i=1}^n \lambda_i$$

If the regression requires gradients with respect to the total modifier Λ_n , then

$$\frac{\partial X}{\partial \Lambda_n} = \frac{\partial X}{\partial (\Lambda_{n-1} \lambda_n)} = \frac{1}{\Lambda_{n-1}} \cdot \frac{\partial X}{\partial \lambda_n}$$

In SimOpt, the regression operates on the parameter Λ_n and the sensitivities provided by Eclipse are divided by Λ_n at each regression iteration. For regression loops such as SimOpt relying on the cumulative parameters Λ_n in this way, it is suggested that the log transform $\lambda = \ln(\lambda)$ is applied.

Eclipse 100

For parameters whose cumulative modifiers are additive (such as HMSWCR and HMSOWCR, in HMxxxxxx), the cumulative modifier would be

$$\Lambda_n = \sum_{i=1}^n \lambda_i$$

in which case no modification is required because

$$\frac{\partial X}{\partial \lambda_n} = \frac{\partial X}{\partial (\lambda_{n-1} + \lambda_n)} = \frac{\partial X}{\partial \lambda_n}$$

For any objective function $G = G(X, \lambda)$, used during regression, then

$$\frac{dG}{d\lambda} = \sum_t \left(\frac{\partial G(X_t, \lambda)}{\partial X_t} \frac{\partial X_t}{\partial \lambda} \right) + \frac{\partial G}{\partial \lambda}$$

so once $\partial X_t / \partial \lambda$ is known then regression can be performed for any objective function G .

Explicit gradients (Eclipse 100)

Where the discretization is not fully implicit, Eclipse accounts for the derivatives of these explicit terms in computing the gradients. At present, this is the case for transmissibility multipliers induced by rock compaction ([ROCKTAB](#), [ROCKTABH](#)) and also for the VE pseudo capillary pressure terms when the vertical equilibrium option is in use.

Instantaneous gradients

The instantaneous gradient facility provides an alternative set of derivatives to those generated for history matching purposes. Recall the history matching philosophy in which derivatives of various solution quantities at the current timestep are computed with respect to variations in a set of parameters at time zero. In contrast, the instantaneous gradient option computes derivatives of solution quantities at the current timestep with respect to variations in the parameters at the current timestep. The instantaneous gradient option harness the existing Eclipse gradient code with the exception that the gradient data need not be propagated forward in time. As a result the instantaneous gradient option can be switched on and off during the course of a simulation via the [GUPFREQ](#) keyword.

The aim of the instantaneous gradient feature is to provide information on the sensitivity of a well to its, and other wells', control modes. For example, if the well P1 is under bottom hole pressure control the gradient of the oil flow rate with respect to the bottom hole pressure can be requested. The calculation of such a gradient combines two factors, the response of the well to a variation in its own bottom hole pressure, and the response of the reservoir to the variation in the well's bottom hole pressure. Since the response of the reservoir is part of the gradient calculation the interaction between wells can also be studied. For example, the relationship between the water flow rate of well P1 and the water injection rate of well I1. However, it should be noted that a number of approximations are made within the gradient calculation that can reduce the accuracy of the calculation when considering the influence of one well upon another.

The instantaneous gradient options is activated via the [GDIMS](#) keyword in the RUNSPEC section. Since the gradient output is available as summary data an unencoded gradient license is required. In the summary section the gradient of the oil, water, gas flow rates and the bottom hole pressure with respect to a gradient parameter can be requested with the keywords WOFWCn, WWFWCn, WGFWCn and WBHWCn, where n is the parameter number required. The gradient parameters are defined by specifying wells using the [GWRTWCV](#) keyword in the SCHEDULE section. Wells selected as gradient parameters must be under either oil, water, gas rate or bottom hole pressure control since the response of the group hierarchy is not included in the gradient calculation. In addition the feature is restricted to the standard well model in unrefined grids.

Example usage

Specify two instantaneous gradient parameters in the RUNSPEC section.

```
GDIMS
2 /
```

Request gradient output in the SUMMARY section. The first item requests that the gradient of the oil flow rate for wells P1 and P2 with respect to the instantaneous gradient parameter 1. The second item that the water flow rate for wells P1 and I1 with respect to the instantaneous gradient parameter 2.

```
WOFWC1
P1 P2 /

WWFWC2
P1 I1 /
```

Specify the instantaneous gradient parameters in the SCHEDULE section. Define the first parameter to be the control mode of well P1, the second to be the control mode of well I1. Compute the gradients at the next timestep only. The values of the requested instantaneous gradient summary mnemonics will be frozen at the last computed value.

```
GWRTWCV
P1 / -- gradient parameter 1
I1 / -- gradient parameter 2
/

GUPFREQ
NEXT /
```

At a later point in the simulation re-specify the instantaneous gradient parameters and compute the gradients for all subsequent timesteps.

```
GWRTWCV
P2 / -- gradient parameter 1
I2 / -- gradient parameter 2
/

GUPFREQ
ALL /
```

Summary of keywords

RUNSPEC section

Keyword	Description
FMTHMD	Declares that the HMD file is to be formatted.
GDIMS	Specifies dimensions for the instantaneous gradient option.
HMDIMS	Specifies dimensions for the gradient option.
RPTHMD	Specifies controls on the output to the HMD file.
UNCODHMD	Requests that the HMD file contains unencoded derivatives.

Parameter specification keywords

GRID section

Keyword	Description
HMFAULTS	Specifies gradients for faults.
HMMULRGT	Specifies inter-region transmissibility parameters.
HMAQUNUM	Specifies gradients for numerical aquifer parameters.

PROPS section

- HMROCK Specifies rock compressibility parameters.
- HMROCKT Specifies rock compaction parameters.

REGIONS section

HMxxxxxx Specifies gradient parameters. The following options apply to the HMxxxxxx keyword.

Restriction	Argument	Description
	HMTRANX	Specifies X-direction transmissibility parameters.
	HMTRANY	Specifies Y-direction transmissibility parameters.
	HMTRANZ	Specifies Z-direction transmissibility parameters.
	HMTRNXY	Specifies X- and Y-direction transmissibility parameters.
	HMTRANR	Specifies R-direction transmissibility parameters.
	HMTRANT	Specifies theta-direction transmissibility parameters.
	HMPORVM	Specifies pore volume parameters.
Eclipse 100	HMSIGMA	Specifies dual porosity sigma factor parameters.
	HMPERMX	Specifies X-direction permeability parameters.
	HMPERMY	Specifies Y-direction permeability parameters.
	HMPRMXY	Specifies XY-direction permeability parameters.
	HMPERMZ	Specifies Z-direction permeability parameters.
	HMTRNX-	Specifies negative X direction transmissibility parameters.
	HMTRNY-	Specifies negative Y-direction transmissibility parameters.
	HMTRNZ-	Specifies negative Z-direction transmissibility parameters.
	HMSWCR	Specifies critical water saturation parameters.
	HMSOWCR	Specifies critical oil-in-water saturation parameters.
	HMSOGCR	Specifies critical oil-in-gas saturation parameters.
	HMSWL	Specifies connate water saturation parameters.
	HMKRW	Specifies maximum water relative permeability parameters.
	HMKRG	Specifies maximum gas relative permeability parameters.
	HMKRO	Specifies maximum oil relative permeability parameters.

Restriction	Argument	Description
	HMKRWR	Specifies maximum displaced water relative permeability parameters.
	HMKRGR	Specifies maximum displaced gas relative permeability parameters.
	HMKRORG	Specifies maximum displaced oil-in-gas relative permeability parameters.
	HMKRORW	Specifies maximum oil-in-water displaced relative permeability parameters.
	HMSWLPC	Specifies lower water saturation for capillary pressure parameter.
	HMSG LPC	Specifies lower gas saturation for capillary pressure parameter.
	HMPCW	Specifies maximum water capillary pressure parameter.
	HMPCG	Specifies maximum gas capillary pressure parameter.

SOLUTION section

Keyword	Description
HMAQUCT	Specifies gradients for Carter-Tracy aquifers.
HMAQUFET	Specifies gradients for Fetkovich aquifers.
HMWELCON	Specifies gradients for well connections

SCHEDULE section

Keyword	Description
GUPFREQ	Specifies update frequency for the instantaneous gradient option.
GWRTWCV	Specifies parameters for the instantaneous gradient option.
HMWPIMLT	Specifies gradients with respect to well productivity index.

Parameter modifier keywords

The cumulative multipliers for gradient parameters are given by the [HMMULTxx](#) keyword. The following table shows which section they can be used in together with the usage of the other parameter modifier keywords.

Section	Keyword	Description
GRID and EDIT	HMMULTFT	Specifies modifiers for fault transmissibilities.
GRID	HMMULTSG	Specifies multipliers on dual porosity sigma factor.
GRID and EDIT	HMMULTPV	Specifies pore volume modifiers.
GRID and EDIT	HMMULTX	Specifies X-direction transmissibility modifiers.
GRID and EDIT	HMMULTY	Specifies Y-direction transmissibility modifiers.
GRID and EDIT	HMMULTXY	Specifies XY-direction transmissibility modifiers.
GRID and EDIT	HMMULTZ	Specifies Z-direction transmissibility modifiers.
GRID and EDIT	HMMULTX-	Specifies negative X-direction transmissibility modifiers.
GRID and EDIT	HMMULTY-	Specifies negative Z-direction transmissibility modifiers.
GRID	HMMREGT	Specifies inter-region transmissibility modifiers.

Section	Keyword	Description
GRID	HMMULTR	Specifies R-direction transmissibility modifiers.
GRID and EDIT	HMMULTTH	Specifies theta-direction transmissibility modifiers.
GRID	HMMLTPX	Specifies X-direction permeability modifiers.
GRID	HMMLTPY	Specifies Y-direction permeability modifiers.
GRID	HMMLTPZ	Specifies Z-direction permeability modifiers.
GRID	HMMLTPXY	Specifies XY-direction permeability modifiers.
EDIT	HMMULTTH	Specifies theta-direction transmissibility modifiers.
GRID	HMMULTSG	Specifies multipliers on dual porosity sigma factor.

PROPS section

[HMPROPS](#) Section heading for property modifiers. The following options apply to the HMPROPS keyword.

Argument	Description
HASWCR	Specifies critical water saturation modifiers.
HASOWCR	Specifies critical oil-in-water saturation modifiers.
HASOGCR	Specifies critical oil-in-gas saturation modifiers.
HASWL	Specifies connate water saturation modifiers.
HMKRW	Specifies maximum water relative permeability modifiers.
HMKRG	Specifies maximum gas relative permeability modifier.
HMKRO	Specifies maximum oil relative permeability modifier.
HMKRWR	Specifies maximum displacing water relative permeability modifier.
HMKRGR	Specifies maximum displacing gas relative permeability modifier.
HMKRORG	Specifies maximum displacing oil-in-gas relative permeability modifier.
HMKRORW	Specifies maximum displacing oil-in-water relative permeability modifier.
HASWLPC	Specifies modifier on lower water saturation used for water capillary pressure.
HASGLPC	Specifies modifier on lower gas saturation used for gas capillary pressure.
HMPCW	Specifies modifier on water capillary pressure.
HMPCG	Specifies modifier on gas capillary pressure.

SOLUTION section

Keyword	Description
HMMLCTAQ	Specifies modifiers for Carter-Tracy aquifers.
HMMLFTAQ	Specifies modifiers for Fetkovich aquifers.
HMMLTWCN	Specifies modifiers for well connections

SUMMARY keywords

Refer to [Simulator performance keywords](#) in the *Eclipse Reference Manual*.

Miscellaneous keywords

Keyword	Description
GRADRESV	Specifies derivatives of solution quantities to be output to the RESTART file.
GRADRFT	Specifies derivatives of RFT quantities to be output to the RFT file.
GRADWELL	Specifies well quantities to be output to the HMD file.
GRADGRUP	Specifies group quantities to be output to the HMD file.
HMPROPS	Flags start of HMPROPS section.
HMRREF	Specifies reference pressures for rock compaction modifiers.
NOHMD	Switch off gradient calculations.
NOHMO	Switch off gradient calculations.
RPTHMG	Configure output of derivatives for groups.
RPTHMW	Configure output of derivatives for wells.
TUNINGH	Tuning parameters for gradient calculations.

Example problem (Eclipse 100)

RUNSPEC

```

TITLE
  HISTORY MATCHING TEST CASE

DIMENS
  10  10  3  /

NONNC

FAULTDIM
  20  /

OIL

WATER

GAS

DISGAS

FIELD

EQLDIMS
  1  100  10  1  1  /

TABDIMS
  2  2  16  12  3  12  1  1  /

REGDIMS
  3  3  0  /

AQUDIMS
  0  0  2  0  2  120  /

HMDIMS
  4  4  10  2  6  /

WELLDIMS
  4  1  1  4  /

```

```

START
  19 'OCT' 1982 /

NSTACK
  24 /

FMTOUT

FMTIN

FMTHMD

```

GRID

In this section, the geometry of the simulation grid and the rock permeabilities and porosities are defined.

The X and Y direction cell sizes (**DX**, **DY**) and the porosities are constant throughout the grid. These are set in the first 3 lines after the **EQUALS** keyword. The cell thicknesses (**DZ**) and permeabilities are then set for each layer. The cell top depths (**TOPS**) are needed only in the top layer (although they could be set throughout the grid). The specified **MULTZ** values act as multipliers on the transmissibilities between the current layer and the layer below.

```

GRID -----
-- ARRAY  VALUE          ----- BOX -----

EQUALS
  'DX'      1000          /
  'DY'      1000          /
  'PORO'    0.3           /

  'DZ'      20           1 10 1 10 1 1 /
  'PERMX'   500          /
  'TOPS'    8325         /

  'DZ'      30           1 10 1 10 2 2 /
  'PERMX'   50           /
  'TOPS'    8345         /

  'DZ'      50           1 10 1 10 3 3 /
  'PERMX'   200          /
  'TOPS'    8395         /

/      EQUALS IS TERMINATED BY A NULL RECORD
-- THE Y AND Z DIRECTION PERMEABILITIES ARE COPIED FROM PERMX
-- SOURCE  DESTINATION  ----- BOX -----

COPY
  'PERMX'   'PERMY'      1 10 1 10 1 3 /
  'PERMX'   'PERMZ'      /

/

FAULTS
  'MAJOR'   3 3 1 4 1 3 'X' /
  'MAJOR'   3 5 4 4 1 3 'Y' /
  'MAJOR'   5 5 5 10 1 3 'X' /
  'MINOR'   6 6 1 3 1 3 'X' /
  'MINOR'   6 10 3 3 1 3 'Y' /

/

MULTFLT
  'MAJOR'   0.001 /
  'MINOR'   01 /

/

HMFaults
  'MAJOR'   /
  'MINOR'   /

/

OLDTRAN
EQUALS

```

```

'HMMULTX' 1.05 2 5 1 10 1 2 /
'HMMULTY' 1.05 1 10 2 5 1 2 /
/
-- OUTPUT OF DX, DY, DZ, PERMX, PERMY, PERMZ, MULTZ, PORO AND TOPS DATA
-- IS REQUESTED, AND OF THE CALCULATED PORE VOLUMES AND X, Y AND Z
-- TRANSMISSIBILITIES

RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTZ'
'PORO' 'TOPS' 'PORV' 'TRANX' 'TRANX' 'TRANX' 'TRANZ' /

```

PROPS

The PROPS section defines the relative permeabilities, capillary pressures, and the PVT properties of the reservoir fluids.

Water relative permeability and capillary pressure are tabulated as a function of water saturation.

```

PROPS-----
      SWAT      KRW      PCOW
SWFN
  0.12  0        0
  1.0   1.0      0 /
/

```

Similarly for gas:

```

-- SGAS      KRG      PCOG
SGFN
  0        0        0
  0.02     0        0
  0.05     0.005    0
  0.12     0.025    0
  0.2      0.075    0
  0.25     0.125    0
  0.3      0.19     0
  0.4      0.41     0
  0.45     0.6      0
  0.5      0.72     0
  0.6      0.87     0
  0.7      0.94     0
  0.88     0.98     0
/
/

```

Oil relative permeability is tabulated against oil saturation for oil-water and oil-gas-concrete water cases:

```

-- SOIL      KROW      KROG
SOF3
  0          0          0
  0.18       0          0
  0.28       0.0001    0.0001
  0.38       0.001     0.001
  0.43       0.01      0.01
  0.48       0.021     0.021
  0.58       0.09      0.09
  0.63       0.2       0.2
  0.68       0.35      0.35
  0.76       0.7       0.7
  0.83       0.98      0.98
  0.86       0.997     0.997
  0.879      1         1
  0.88       1         1 /
/

```

PVT properties of water

```
-- REF. PRES. REF. FVF COMPRESSIBILITY REF VISCOSITY VISCOSIBILITY
PVTW      4014.7      1.029      3.13D-6      0.31      0 /
/
```

Rock compressibility

```
-- REF. PRES COMPRESSIBILITY
ROCK      14.7      3.0D-6      /
/
```

Surface densities of reservoir fluids

```
-- OIL WATER GAS
DENSITY   49.1  64.79  0.06054 /
/
```

PVT properties of dry gas with no vaporized oil (use [PVTG](#) to specify the properties of Wet Gas)

```
-- PGAS BGAS VISGAS
PVDG
  14.7 166.666 0.008
 264.7 12.093 0.0096
 514.7  6.274 0.0112
1014.7  3.197 0.014
2014.7  1.614 0.0189
2514.7  1.294 0.0208
3014.7  1.080 0.0228
4014.7  0.811 0.0268
5014.7  0.649 0.0309
9014.7  0.386 0.047 /
/
```

PVT properties of live oil with dissolved gas (use [PVDO](#) to specify the properties of dead oil).

For each value of R_s the saturation pressure, FVF and viscosity are specified. For $R_s=1.27$ and 1.618, the FVF and viscosity of Undersaturated Oil are defined as a R_s , but must be supplied for the highest R_s (1.618).

```
-- RS POIL FVFO VISO
PVTO
  0.001  14.7 1.062 1.04 /
  0.0905 264.7 1.15 0.975 /
  0.18  514.7 1.207 0.91 /
  0.371 1014.7 1.295 0.83 /
  0.636 2014.7 1.435 0.695 /
  0.775 2514.7 1.5 0.641 /
  0.93  3014.7 1.565 0.594 /
  1.270 4014.7 1.695 0.51
        5014.7 1.671 0.549
        9014.7 1.579 0.74 /
  1.618 5014.7 1.827 0.449
        9014.7 1.726 0.605 /
  1.630 8014.7 1.828 0.449
        9014.7 1.8 0.605 /
/
/
```

Carter Tracy aquifer influence table

```
AQUTAB
  0.01  0.112
  0.05  0.229
```

```

0.1 0.315
0.15 0.376
0.2 0.424
0.25 0.469
0.3 0.503
0.4 0.564
0.5 0.616
0.6 0.659
/

```

Output controls for PROPS data activated for **SOF3**, **SWFN**, **SGFN**, **PVTW**, **PVDG**, **DENSITY** and **ROCK** keywords

```

RPTPROPS
'SOF2' '=' 1 'SWFN' '=' 1 'SGFN' '=' 1 'PVTW' '=' 1
'PVTG' '=' 1 'DENSITY' 'ROCK' /

```

REGIONS

The REGIONS section defines how the reservoir is split into regions by saturation function, PVT function, fluid in place region and gradient region for example.

```

REGIONS -----
EQUALS
'PVTNUM' 2 1 10 1 10 1 3 /
'SATNUM' 2 1 10 1 10 1 3 /
'HMTRANZ' 1 1 10 1 10 1 3 /
'HMTRANX' 1 1 10 1 10 1 3 /
'HMTRNXY' 1 1 10 1 10 1 3 /
'HMTRANY' 1 1 10 1 10 1 1 /
'HMTRANY' 2 1 10 1 10 1 2 /
'HMTRANY' 3 1 10 1 10 1 3 /
/

BOX
1 10 1 10 2 2 /

SATNUM
1 2 1 2 1 2 1 2 1 2
2 1 2 1 2 1 2 1 2 1
1 2 1 2 1 2 1 2 1 2
2 1 2 1 2 1 2 1 2 1
1 2 1 2 1 2 1 2 1 2
2 1 2 1 2 1 2 1 2 1
1 2 1 2 1 2 1 2 1 2
2 1 2 1 2 1 2 1 2 1
1 2 1 2 1 2 1 2 1 2
2 1 2 1 2 1 2 1 2 1
/

ENDBOX

COPY
'SATNUM' 'FIPXXX' /
/

EQUALS
'FIPNUM' 1 4* 1 1 /
'FIPNUM' 2 4* 2 2 /
'FIPNUM' 3 4* 3 3 /
'FIPYYY' 1 1 3 1 10 1 3 /
'FIPYYY' 2 4 7 4* /
'FIPYYY' 3 7 10 4* /
/

RPTREGS
'PVTNUM' 'SATNUM' 'EQLNUM' 'FIPNUM' 'HM' /

```

SOLUTION

The SOLUTION section defines the initial state of the solution variables (phase pressures, saturations and Gas-Oil ratios)

```

SOLUTION -----
-- DATA FOR INITIALISING FLUIDS TO POTENTIAL EQUILIBRIUM
--
--      DATUM   DATUM   OWC   OWC   GOC   GOC   RSVD   RVVD   SOLN
--      DEPTH   PRESS   DEPTH  PCOW  DEPTH  PCOG  TABLE TABLE  METH
EQUIL
      8400    4800    8500     0    8200     0       1       0       0 /
-- VARIATION OF INITIAL RS WITH DEPTH
--
--      DEPTH    RS
RSVD
      8200    1.270
      8500    1.270 /
--CARTER TRACY AQUIFER CONNECTING TO BOTTOM FACE OF RESERVOIR
AQUCT
  1  5100.0  1*  20.0   0.10  3.0E-5  2400.0  140.0  360  1  1  /
/
AQUANCON
  1  1  10   1  10   3   3  'K+' /
/
HMAQUCT
--      PERM    THETA
--  1  'YES'    'YES' /
/
-- OUTPUT CONTROLS
-- (SWITCH ON OUTPUT OF INITIAL GRID BLOCK PRESSURES
--                               FLUID VISCOSITIES
--                               PHASE RESERVOIR DENSITIES )
RPTSOL
  'PRES'  'FIP=3'  'VOIL'  'VWAT'  'VGAS'  'DENO'
  'DENW'  'DENG'  /

```

SUMMARY

This section specifies data to be written to the SUMMARY files and, which may later be used with the Eclipse graphics package

```

-- FIELD OIL PRODUCTION
FOPR
-- WELL GAS-OIL RATIO FOR PRODUCER

WGOR
'PRODUCER'
/
-- WELL BOTTOM-HOLE PRESSURE

WBHP
'PRODUCER'
/
--AQUIFER INFLUX

FAQT
AAQR
/
AAQR
/

```

```
-- CPU TIMES

TCPU

TCPUH
/
```

SCHEDULE

The SCHEDULE section defines the operations to be simulated

```
SCHEDULE -----
RPTSCHED
  'PRES'  'SOIL'  'SWAT'  'SGAS'  'RS'   'FIP=3'
  'WELLS' 'CPU=2' 'WELSPECS' 'NEWTON=2'
  'VOIL'  'VWAT'  'VGAS'  'DENO'  'DENW'  'DENG'  /

RPTRST
  'BASIC=3'  'FIP'  'FREQ=2'  'PRES'  'VISC'
  'DEN'      /

DRSDT
  100.0
/

GRADWELL
  'WBHP' /
  'WBP'  /
  'WWCT' /
  'WOPR' /
  'WWPR' /
/

RPTHMW
  'PROD*' 'ON' /
  'INJE*' 'OFF' /
/
```

Set initial timestep to 1 day and maximum to 6 months

```
TUNING
/
/
8* 200.0 /
```

Well specification data

```
-- WELL   GROUP LOCATION BHP   PI
-- NAME      NAME    I  J     DEPTH DEFN

WELSPECS
  'PRODUCER' 'G'    10 10     8400 'OIL' /
  'PRODUCE2' 'G'    10 1     8400 'OIL' /
  'INJECTOR'  'G'     1 1     8335 'GAS' /
  'INJECT2'   'G'     1 10    8335 'GAS' /
/
```

Completion specification data

```
-- WELL   -LOCATION- OPEN/ SAT CONN WELL
-- NAME      I  J K1 K2 SHUT  TAB FACT DIAM

COMPDAT
  'PRODUCER' 10 10 3 3 'OPEN' 0 -1 0.5 /
  'PRODUCE2' 10 1 3 3 'OPEN' 0 -1 0.5 /
  'INJECTOR'  1 1 1 1 'OPEN' 1 -1 0.5 /
  'INJECT2'   1 10 1 1 'OPEN' 1 -1 0.5 /
```

```

/
WCONHIST
  'PRODUCER' 'OPEN' 'ORAT' 20000 /
  'PRODUCE2' 'OPEN' 'ORAT' 20000 /
/

WELTARG
  'PRODUCER' 'BHP' 1000 /
  'PRODUCE2' 'BHP' 1000 /
/

```

Injection well controls

```

--      WELL      INJ  OPEN/  CNTL   FLOW
--      NAME      TYPE  SHUT   MODE   RATE      BHP

WCONINJE
  'INJECTOR' 'GAS' 'OPEN' 'RATE' 100000 1* 10000 /
  'INJECT2' 'GAS' 'OPEN' 'RATE' 100000 1* 10000 /
/

TSTEP
6*40
/

END

```

Eclipse 300

Note: The above example shows the majority of parameters available for use within Eclipse 100. At present the only parameters available for use within Eclipse 300 are pore volume, transmissibility and permeability multipliers.

Adjoint gradient screening

	Eclipse 100
x	Eclipse 300

This chapter describes the adjoint gradient screening facilities within Eclipse 300.

For analysis of simulation models, methods that use gradient information can be prohibitively expensive if the gradients are derived numerically, from multiple runs of the 'black box'.

Eclipse production profile gradients can be calculated much more efficiently from within the simulator itself (see "[Gradient option](#)"). The SimOpt interactive application successfully makes use of these Eclipse gradients in an external gradient-based optimization loop, as an aid in the history matching phase of a study. With the classic type of gradient calculation used in the SimOpt workflow, there is still a practical limit on the number of parameters that can be used, as the simulation run time increases as the number of parameters increases.

An alternative gradient calculation approach, the adjoint method, allows cost-effective calculation of sensitivities of an objective function with respect to a large number of static grid parameters relevant to the flow equations, such as porosity or permeability of every active grid block. Here, an objective function is defined as a dynamic flow-related parameter (component rate, flowing bottom-hole pressure) of a particular well. The adjoint gradient method for static parameters is available for use.

The adjoint method for gradient calculation is also used in the optimization of well control settings to maximize production from a simulation model (see "[Optimization](#)").

The adjoint production optimization method cannot be used at the same time as this adjoint gradient screening option.

The sections that follow give an overview of the keywords necessary to use this option. The output from this option is described as well as a list of the restrictions and limitations of its use. A description of the formulation is also provided.

Keywords

In the RUNSPEC section, the [AJGRADNT](#) keyword must be specified to activate the option.

Specifying the objective functions

In the SCHEDULE section, the required objective functions are specified using the [AJGWELLS](#) keyword, which consists of a list of well names and mnemonics for dynamic parameters.

If an adjoint gradient screening run terminates early (for instance, because all the wells are shut, or because an economic limit has been violated) and this keyword has not been specified by the early exit time, then no adjoint gradients will be generated as none will have been requested by that stage of the simulation. In order to avoid this problem, the [AJGWELLS](#) keyword should be specified right at the beginning of the SCHEDULE section.

Note: This advice on the positioning of the [AJGWELLS](#) keyword is different to the advice given pre-2007.1. Previously it was recommended that this keyword should be placed just before the [END](#) keyword.

Specifying the parameters

Also in the SCHEDULE section, the required static properties are specified using the [AJGPARAM](#) keyword. The sensitivities will be calculated with respect to these properties on each simulation grid cell.

If an adjoint gradient screening run terminates early (for instance, because all the wells are shut, or because an economic limit has been violated) and this keyword has not been specified by the early exit time, then no adjoint gradients will be generated as none will have been requested by that stage of the simulation. In order to avoid this problem, the `AJGPARAM` keyword should be specified right at the beginning of the `SCHEDULE` section.

Example

```

SCHEDULE

AJGPARAM
'PERMX' 'PORO' /

AJGWELLS
'P1' 'WBHP' /
'P2' 'WBHP' /
'I1' 'WBHP' /
/

RPTRST
BASIC=2 /

WELSPECS
'P1' 'GRUP' 1 1 3000.0 'OIL' 1* 'NO' /
'P2' 'GRUP' 1 5 3000.0 'OIL' 1* 'NO' /
'I1' 'GRUP' 5 1 3000.0 'WATER' 1* 'NO' /
/

COMPDAT
'P1' 1 1 1 1 'OPEN' 0 1* 0.2 1* 0.0 1* 'Z' /
'P2' 1 5 1 1 'OPEN' 0 1* 0.2 1* 0.0 1* 'Z' /
'I1' 5 1 1 1 'OPEN' 0 1* 0.2 1* 0.0 1* 'Z' /
/

WCONPROD
'P*' 'OPEN' 'LRAT' 3* 100 1* 1* /
/

WCONINJE
'I1' 'WAT' 'OPEN' 'RATE' 400 1* 1*
/

TSTEP
50*5.0 /

END

```

Output

In the `SCHEDULE` section, the `RPTRST` keyword must be specified with `BASIC=2` or `BASIC=6` so that the restart information is saved for every report/timestep.

At the end of the simulation, the sensitivities are calculated for all objective functions and at every report step. These are stored as real arrays in an additional section at the end of each restart file. The sensitivities can be plotted by loading restart data into FloViz, Petrel or other visualization packages, and then displaying `AJGFN` restart properties.

Note: The sensitivities output to the restart file will be encoded unless the `UNCODHMD` keyword is present in the `RUNSPEC` section. The encoded sensitivities are not decoded by Eclipse Office or FloViz and so are not meaningful when visualized in those programs.

Example

For the example in "Keywords", which has 25 simulation grid cells, the section added to the restart file for report step 49 will be as shown:

```
'AJGRADNT'          0 'MESS'
'AJGNAMES'          24 'CHAR'
'AJGFN  1' 'P1'      'WBHP'      'PERMX' 'AJGFN  2' 'P2'      'WBHP'
'PERMX' 'AJGFN  3' 'I1'      'WBHP'      'PERMX' 'AJGFN  4' 'P1'
'WBHP' 'PORO' 'AJGFN  5' 'P2'      'WBHP' 'PORO' 'AJGFN  6'
'I1' 'WBHP' 'PORO'
'AJGPARA'          5 'INTE'
49          1          1          25          0
'AJGFN  1'          25 'REAL'
0.26733625E+00  0.28940764E+00  0.15707164E+00  0.82344688E-01
0.53730132E-02 -0.60202714E-01 -0.42874895E-01 -0.30267414E-01
-0.30209363E-01 -0.54189377E-01 -0.10426258E-01 -0.54382261E-01
-0.51501323E-01 -0.38832009E-01 -0.45974124E-01  0.63967048E-02
-0.14613148E-01 -0.23543639E-01 -0.24201822E-01 -0.31580176E-01
-0.64227316E-02 -0.27032575E-01 -0.31063568E-01 -0.24502732E-01
-0.17454501E-01
'AJGPARA'          5 'INTE'
49          2          1          25          0
'AJGFN  2'          25 'REAL'
0.12758227E+00  0.28352138E+00  0.15206486E+00  0.75946450E-01
0.50332956E-02 -0.70431128E-01 -0.43791033E-01 -0.33293936E-01
-0.32356218E-01 -0.46265345E-01  0.21185523E-01 -0.51040705E-01
-0.59991647E-01 -0.43821692E-01 -0.33047505E-01  0.98482676E-01
-0.37209673E-02 -0.28060049E-01 -0.27350528E-01 -0.15706640E-01
0.32918173E+00  0.45060366E-01  0.36429397E-02 -0.28943263E-02
-0.19188058E-02
...
Lines skipped here
...
'ENDAJG'          0 'MESS'
```

The new records appended to the restart file are:

- AJGRADNT - the header marking the start of the section
- AJGNAMES - a character array consisting of blocks of four fields containing the following information
 - Name of the sensitivity (internally assigned by Eclipse)
 - Name of the well
 - Name of the dynamic parameter on the well
 - Name of the static grid property
- AJGPARA - an integer array containing the following information for the record that appears next in the file
 - Report step number
 - Well number
 - Parameter number
 - Number of sensitivity values that will be output
 - Reserved for future use
- AJGFN 1 - a real array containing one sensitivity value for every active grid cell. Further arrays are numbered sequentially AJGFN 2, AJGFN 3 and so on.

- ENDAJG - the header marking the end of the adjoint gradient section.

Restrictions

This section lists the current restrictions and known limitations of the option.

- This option is only available in Eclipse 300.
- This option cannot be used with the [PARALLEL](#) option.
- This option is not compatible with [IMPES](#), [IMPSAT](#) or [AIM](#) solution methods. The [FULLIMP](#) solution method must be used.
- This option cannot be used with reservoir optimization (see the [RESOPT](#) keyword).
- The current version requires multiple restart files format (see the [MULTOUT](#) keyword). Also, restart information must be saved at every report step using the BASIC=2 or BASIC=6 options in [RPTRST](#) keyword.
- In order for visualization of adjoint gradient properties to be available in post-processing programs such as Eclipse Office Results Viewer, FloViz or Petrel, the current version of Eclipse 300 requires that a restart index file (which has an RSSPEC extension) is **not** written out. For this reason, the presence of the [AJGRADNT](#) keyword automatically turns on [NORSSPEC](#).
- The option does not allow adjoint gradient calculations for wells under reservoir voidage rate or THP control, or wells under active group control (see the [WCONPROD](#) keyword).
- Simulation grids with non-neighbor connections cannot be used (for instance, simulation models containing faults, numerical aquifers, local refinement or coarsening, and dual-porosity). It is advisable to explicitly use [NONNC](#) keyword in the RUNSPEC section.
- Any well used for objective function definition must be on a surface rate or bottom-hole pressure control during the entire simulation run. When a requested dynamic parameter is used as well control, or the well is shut or stopped, the derivative is set to zero.

Formulation

Eclipse forces the fully implicit residual in each simulator grid block, R , to be zero (see "[Formulation of the equations](#)") by solving for solution variables $X(t)$ at discrete points in time, $t_1, t_2, \dots, t_N$, so that for a dynamic well parameter, the following vector notation can be used.

$$W[X(t)] \rightarrow \underline{W} = \begin{bmatrix} W[X(t_1)] \\ W[X(t_2)] \\ \dots \\ W[X(t_N)] \end{bmatrix} \equiv \begin{bmatrix} W_1 \\ W_2 \\ \dots \\ W_N \end{bmatrix}$$

Assuming that a parameter p , which is normally a constant in the flow equations, can be varied, then $W = W(p, X(t, p))$ which has the following differential:

$$dW = \left(\frac{\partial W}{\partial p} \right) \cdot dp + \left(\frac{\partial W}{\partial X} \right) \cdot dX \quad \text{Eq. 13.29}$$

and that the flow equations still have to be honored:

$$\begin{aligned}
dR_1(X_1, X_0, p) &= 0 \\
dR_2(X_2, X_1, p) &= 0 \\
dR_3(X_3, X_2, p) &= 0 \\
&\dots \\
dR_N(X_N, X_{N-1}, p) &= 0
\end{aligned}
\tag{Eq. 13.30}$$

then the term containing solution variables in equation 13.29 can be eliminated. To do this, apply Lagrange multipliers, Ψ , for each equation in equation 13.30, and insert them into equation 13.29 to give the following:

$$\frac{dW}{dp} = \left\{ \left(\frac{\partial W}{\partial p} \right) + \Psi \cdot \left(\frac{\partial R}{\partial p} \right) \right\} + \left\{ \left(\frac{\partial W}{\partial X} \right) + \Psi \cdot \left(\frac{\partial R}{\partial X} \right) \right\} \cdot \frac{dX}{dp}
\tag{Eq. 13.31}$$

Then choose the Lagrange multipliers so that the right hand term in equation 13.31 is equal to zero - this choice has the benefit that there is no need to calculate the (expensive) dX/dp term.

These Lagrange multipliers can be represented as a lower triangular matrix:

$$\Psi = ()$$

where both row and column indices indicate the report step number.

So, at each time level, k , the Lagrange multipliers can be calculated by solving the “adjoint” equations, equations 13.32 and 13.33:

$$\frac{\partial W_k}{\partial X_k} + \Psi_{k,k} \cdot \frac{\partial R_k}{\partial X_k} = 0
\tag{Eq. 13.32}$$

$$\Psi_{k,m} \cdot \frac{\partial R_m}{\partial X_m} + \Psi_{k,m+1} \cdot \frac{\partial R_{m+1}}{\partial X_m} = 0
\tag{Eq. 13.33}$$

where $m = 1, \dots, k-1$.

The residual derivatives required in this system are the same as the ones already used in the non-linear iterations inside the simulator, so they are already calculated and available for this method to use.

Note: With this adjoint method, the number of equations that must be solved grows quadratically with the number of report steps. However, for a given report step, the number of parameters are not limiting as the residual derivatives required in equation 13.31 are relatively cheap to obtain.

Petro-elastic model

x	Eclipse 100
	Eclipse 300

The petro-elastic model option provides the ability to generate simulated seismic-derived quantities (such as acoustic impedance) in each active grid cell.

Note: You must have an `eclipse4d` license in order to access this feature. See "[Licenses](#)" for more information.

The acoustic response of the reservoir is a combination of the acoustic responses of the different rocks and fluids that constitute the reservoir.

In the petro-elastic model, conventional simulation properties (such as pressure, fluid density, fluid saturations and effective porosity) are used to calculate the effective acoustic response of the fluids. This is then combined with the acoustic response of the rocks (both frame and minerals) to give the overall acoustic response of the reservoir.

The acoustic response is calculated and output for each grid cell in the simulation model.

Assumptions

Gassmann fluid substitution equation

The Petro-elastic Model uses the Gassmann fluid substitution equation.

The use of this equation relies on the following assumptions:

- The seismic frequencies used are sufficiently low (less than 100 Hz) such that the induced pore pressures are equilibrated throughout the pore space (that is there is sufficient time for the pore volume fluid to flow and eliminate wave-induced pore pressure gradients).
- There is a homogeneous mineral modulus.
- The rock is isotropic.
- The fluid bearing rock is completely saturated.

Fluid viscosity

The fluids are assumed to be not "too" viscous so that the shear stiffness of the reservoir is entirely due to the shear stiffness of the dry rock frame.

The petro-elastic model

Most of the theory of this model (that is the assumptions and equations) can be found in [Mavko et al \(1998\)](#).

The petro-elastic model can be decomposed into four levels. These are:

- "Level 1: Impedances and velocities"
- "Level 2: Effective bulk/shear modulus and density"
- "Level 3: Rock properties and fluid properties"
- "Level 4: Individual fluid phase properties"

Level 1 is the output and it requires results from the calculations in Level 2. In turn, Level 2 requires the results from the calculations in Level 3 and so on.

Level 1: Impedances and velocities

The main outputs of the petro-elastic model are:

I_P the impedance to pressure waves for a particular simulation grid cell

I_S the impedance to shear waves for a particular simulation grid cell

ν the Poisson's ratio for a particular simulation grid cell

These are defined as

$$I_P = \rho V_P \quad \text{Eq. 13.34}$$

$$I_S = \rho V_S \quad \text{Eq. 13.35}$$

$$\nu = \frac{\frac{1}{2}V_P^2 - V_S^2}{V_P^2 - V_S^2} \quad \text{Eq. 13.36}$$

where

ρ is the effective density of all the material (fluids, minerals, frame) in the grid cell

V_P is the effective pressure wave velocity in the simulation grid cell

V_S is the effective shear wave velocity in the simulation grid cell

These effective velocities are calculated by

$$V_P = \sqrt{\frac{K + (4G/3)}{\rho}} \quad \text{Eq. 13.37}$$

$$V_S = \sqrt{\frac{G}{\rho}} \quad \text{Eq. 13.38}$$

where

K is the effective bulk modulus of the simulation grid cell

G is the effective shear modulus of the simulation grid cell

Level 2: Effective bulk/shear modulus and density

Effective bulk modulus

The effective bulk modulus of a simulation grid cell is a combination of the effects of the fluids, the frame and the minerals (for example, sand, shale, silt) in the grid cell. The combination is performed using the Gassmann fluid substitution equation. This relies on the assumptions listed in "[Assumptions](#)".

The following form of Gassmann's equation is used in the petro-elastic model:

$$K = K_{frame} + \frac{\left(1 - \frac{K_{frame}}{K_{minerals}}\right)^2}{\frac{\phi_{eff}}{K_{fluid}} + \frac{(1 - \phi_{eff})}{K_{minerals}} - \frac{K_{frame}}{(K_{minerals})^2}} \quad \text{Eq. 13.39}$$

where

K_{frame} is the effective bulk modulus of the dry rock frame for this grid cell

$K_{minerals}$ is the effective bulk modulus of the minerals in this grid cell

K_{fluid} is the effective bulk modulus of the pore fluid in this grid cell

ϕ_{eff} is the effective porosity of this grid cell

Effective shear modulus

With the assumption that the fluids in the grid cell are not too viscous, then the overall shear modulus is just the shear modulus of the dry rock frame, that is:

$$G = G_{frame} \quad \text{Eq. 13.40}$$

where

G_{frame} is the shear modulus of the dry rock frame for this grid cell

Effective density

The effective density in a grid cell is given by the effective porosity weighted average of the fluid density and the mineral density in that grid cell, that is:

$$\rho = \phi_{eff} \cdot \rho_{fluid} + (1 - \phi_{eff}) \cdot \rho_{minerals} \quad \text{Eq. 13.41}$$

where

$\rho_{minerals}$ is the effective density of the minerals in this grid cell

ρ_{fluid} is the effective density of the fluids in this grid cell

The effective density of the minerals is supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

Level 3: Rock properties and fluid properties

Effective porosity

The calculation of the effective porosity for a grid cell is based on the porosity at the reference pressure, ϕ_0 , in the grid cell and the effects of rock compressibility or rock compaction (depending on which type of pressure effect is present in the model).

Rock compressibility

If the simulation model contains the [ROCK](#) keyword, then rock compressibility is used in the calculation of the effective porosity.

$$\phi_{eff} = \phi_0 \left(1 + X + \frac{X^2}{2} \right) \quad \text{Eq. 13.42}$$

where

$$X = C(P - P_{ref}) \quad \text{Eq. 13.43}$$

and

C is the rock compressibility for the grid cell

P is the pressure in the grid cell

P_{ref} is the rock compressibility reference pressure for the grid cell

In each grid cell, the porosity at the reference pressure, and the pressure itself, are available within the simulator. The rock compressibility and its reference pressure are also available as they are specified in the ROCK keyword.

Rock compaction

If the simulation model contains the [ROCKCOMP](#) keyword, then rock compaction is used in the calculation of the effective porosity.

$$\phi_{eff} = \phi_0 C(P)$$

where

$C(P)$ is the rock compaction pore volume multiplier for the grid cell (this is specified as tables of multipliers against pressure in the [ROCKTAB](#) keyword)

P is the pressure in the grid cell

Effective fluid density

The effective fluid density is the saturation weighted average of the individual fluid component densities, that is,

$$\rho_{fluid} = S_o \rho_o + S_w \rho_w + S_g \rho_g \quad \text{Eq. 13.44}$$

where

$S_{o/w/g}$ is the saturation of oil/water/gas in the grid cell

$\rho_{o/w/g}$ is the density of oil/water/gas in the grid cell

These quantities are all available within the simulator for each grid cell.

Effective bulk modulus of the fluid

Two methods have been implemented for calculating the effective bulk modulus of the fluid.

The choice of which method to use is supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

Generalized Wood's Law

This is simply a saturation weighted harmonic average of the individual fluid bulk moduli, that is:

$$\frac{1}{K_{fluid}} = \frac{S_o}{K_o} + \frac{S_w}{K_w} + \frac{S_g}{K_g} \quad \text{Eq. 13.45}$$

where

$K_{o/w/g}$ is the bulk modulus of the oil/water/gas in the grid cell

Wood's Law with Brie et al

This uses a saturation weighted harmonic average for the liquid phases (that is, oil and water) with a separate equation used to incorporate the effect of the gas phase, that is:

$$K_{fluid} = (K_{liquid} - K_g)(1 - S_g)^e + K_g \quad \text{Eq. 13.46}$$

where

e is the “patchy saturation” exponent

K_{liquid} is the combined bulk modulus of the liquid phases, given by

$$\frac{S_o + S_w}{K_{liquid}} = \frac{S_o}{K_o} + \frac{S_w}{K_w} \quad \text{Eq. 13.47}$$

The patchy saturation exponent is supplied directly as an input for each petro-elastic region as part of the Petro-elastic Model definition in the simulation model. This equation was first presented in [Brie et al \(1995\)](#).

Effective bulk modulus of the minerals

A linear function of effective porosity is used to specify the effective bulk modulus of the minerals in the grid cell, that is

$$K_{minerals} = C_1 \phi_{eff} + C_0 \quad \text{Eq. 13.48}$$

where

C_0 is the zeroth order coefficient of effective porosity

C_1 is the first order coefficient of effective porosity

The zeroth and first order coefficients of effective porosity are supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

Effective shear modulus of the minerals

The effective shear modulus of the minerals, $G_{minerals}$, is supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

Effective bulk modulus of the frame

The effective bulk modulus of the dry rock frame is given by:

$$(K_{frame})^\alpha = C_K(K_{minerals})^\alpha + \sum_{i=1}^7 \{C_{K,i}(P) \cdot \phi_{eff}^i\} \quad \text{Eq. 13.49}$$

where

α is a specified exponent

C_K is the coefficient of the $K_{minerals}$ term in the calculation of K_{frame}

$C_{K,i}(P)$ are specified tabular functions of pressure

α is usually -1 or +1 and specifies whether the dry rock frame and mineral bulk moduli are related harmonically or arithmetically.

Both α and the tabular functions are supplied directly as an input for each petro-elastic region as part of the Petro-elastic Model definition in the simulation model.

Effective shear modulus of the frame

The effective shear modulus of the dry rock frame is given by

$$(G_{frame})^\alpha = C_G(G_{minerals})^\alpha + \sum_{i=1}^7 \{C_{G,i}(P) \cdot \phi_{eff}^i\} \quad \text{Eq. 13.50}$$

where

α is the same exponent as specified in equation 13.49

C_G is the coefficient of the $G_{minerals}$ term in the calculation of G_{frame}

$C_{G,i}(P)$ are specified tabular functions of pressure

Both α and the tabular functions are supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

Level 4: Individual fluid phase properties

The calculations for the bulk moduli of the oil, water and gas in a grid cell are based on the empirical equations given in [Mavko et al \(1998\)](#).

Note: In the descriptions of the empirical equations in this section, the relevant units for physical quantities are given. These units are only used for internal calculation purposes. You do not have to input your petro-elastic property data in these units.

Bulk modulus of oil

The bulk modulus of oil in a grid cell is given by

$$K_o = \rho_o V_o^2 \quad \text{Eq. 13.51}$$

where

ρ_o is the density of oil (gm/cc) in the grid cell

V_o is the velocity of sound in oil in the grid cell given (in m/s) by

$$V_o = 2096 \sqrt{\frac{\rho_{o,R}}{2.6 - \rho_{o,R}}} - (3.7 \cdot T) + (4.64 \cdot P) + 0.0115 \cdot \left\{ \left(4.12 \sqrt{\frac{1.08}{\rho_{o,R}}} - 1 \right) - 1 \right\} \cdot T \cdot P \quad \text{Eq. 13.52}$$

where

T is the temperature (°C) in the grid cell. This is supplied directly as an input for each petro-elastic region as part of the petro-elastic model definition in the simulation model.

P is the pressure (MPa) in the grid cell

$\rho_{o,R}$ is the density of oil (gm/cc) in the grid cell at reference conditions (15.6°C and atmospheric pressure). This is calculated by first correction for temperature effects

$$\rho_{o,P} = \rho_o \{ 0.972 + 3.81 \times 10^{-4} (T + 17.78)^{1.175} \} \quad \text{Eq. 13.53}$$

and then correcting for pressure effects using

$$\rho_{o,P} = \rho_{o,R} + (0.00277P - 1.71 \times 10^{-7}P^3)(\rho_{o,R} - 1.15)^2 + 3.49 \times 10^{-4}P$$

Bulk modulus of water

The bulk modulus of water in a grid cell is given by

$$K_w = \rho_w V_w^2 \quad \text{Eq. 13.54}$$

where

V_w is the velocity of sound waves in water in the grid cell given (in m/s) by

$$V_w = V_{pw} + C_S \cdot S + (780 - 10P + 0.16P^2) \cdot S^{1.5} - 1820S^2 \quad \text{Eq. 13.55}$$

where

S is the salinity of the water in the grid cell (in ppm/10⁶, that is, as a fraction of 1)

V_{pw} is the velocity of sound waves in pure water in the grid cell

Again, as this is an empirical formula, P must be in MPa. S is supplied directly as an input for each petro-elastic region as part of the Petro-elastic Model definition in the simulation model, while C_S and V_{pw} are given by

$$C_S = 1170 - 9.6T + 0.055T^2 - 8.5 \times 10^{-5}T^3 + 2.6P - 0.0029TP - 0.0476P^2$$

where

T is in °C, and

$$V_{pw} = \sum_{i=1}^4 \left\{ \sum_{j=0}^3 (w_{ij} T^j P^j) \right\} \quad \text{Eq. 13.56}$$

where

$$\begin{aligned}
w_{00} &= 1402.85 & w_{02} &= 3.437 \times 10^{-3} \\
w_{10} &= 4.871 & w_{12} &= 1.739 \times 10^{-4} \\
w_{20} &= -0.04783 & w_{22} &= -2.135 \times 10^{-6} \\
w_{30} &= 1.487 \times 10^{-4} & w_{32} &= -1.455 \times 10^{-8} \\
w_{40} &= -2.197 \times 10^{-7} & w_{42} &= 5.230 \times 10^{-11} \\
w_{01} &= 1.524 & w_{03} &= -1.197 \times 10^{-5} \\
w_{11} &= -0.0111 & w_{13} &= -1.628 \times 10^{-6} \\
w_{21} &= 2.747 \times 10^{-4} & w_{23} &= 1.237 \times 10^{-8} \\
w_{31} &= -6.503 \times 10^{-7} & w_{33} &= 1.327 \times 10^{-10} \\
w_{41} &= 7.987 \times 10^{-10} & w_{34} &= -4.614 \times 10^{-13}
\end{aligned}$$

Eq. 13.57

Bulk modulus of gas

The bulk modulus of gas in a grid cell is given (in MPa) by

$$K_g = \frac{P\gamma}{1 - \left(\frac{P_r}{Z} \cdot f\right)} \quad \text{Eq. 13.58}$$

where

$$GRAV = \frac{\rho_{gas}|_{surface}}{\rho_{air}|_{surface}}$$

$$P_r = \frac{P}{4.892 - (0.4048 \cdot GRAV)}$$

$$T_r = \frac{T + 273.15}{94.72 + (170.75 \cdot GRAV)}$$

$$f = c \cdot d \cdot m + a$$

$$Z = (a \cdot P_r) + b + (c \cdot d)$$

$$a = 0.03 + 0.00527(3.5 - T_r)^3$$

$$b = 0.642T_r - 0.007T_r^4 - 0.52$$

$$c = 0.109(3.85 - T_r)^2$$

$$d = e^{\left\{ - \left[0.45 + 8 \left(0.56 - \frac{1}{T_r} \right)^2 \right] \frac{P_r^{1.2}}{T_r} \right\}}$$

$$m = -1.2 \left\{ 0.45 + 8 \left(0.56 - \frac{1}{T_r} \right)^2 \right\} \frac{P_r^{0.2}}{T_r}$$

$$\gamma = 0.85 + \frac{5.6}{P_r + 2} + \frac{27.1}{(P_r + 3.5)^2} - 8.7e^{\{-0.65(P_r + 1)\}}$$

GRAV is available within the simulator.

Again, since this is an empirical formula, P is in MPa, and T is in °C.

Implementation of the petro-elastic model

Inputs

The following values, which are not normally present in the simulation model, are required to specify the Petro-elastic Model:

These are specified as constants on each petro-elastic region

S	the salinity of the water in equation 13.55
T	the temperature in equation 13.52, equation 13.55, and equation 13.58 (this may be available elsewhere in the simulation model but the petro-elastic temperature is specified separately)
$\rho_{minerals}$	the effective density of the minerals in equation 13.41
C_1	the first order coefficient of effective porosity in the calculation of the effective bulk modulus of the minerals in equation 13.48
C_0	the zeroth order coefficient of effective porosity in the calculation of the effective bulk modulus of the minerals in equation 13.48
C_K	is the coefficient of the $K_{minerals}$ term in the calculation of K_{frame} in equation 13.49
C_G	the coefficient of the $G_{minerals}$ term in the calculation of G_{frame} in equation 13.50
$G_{minerals}$	the effective shear modulus of the minerals in equation 13.50
α	the exponent in the calculation of the effective bulk/shear modulus of the dry rock frame in equation 13.49 and 13.50
e	the "patchy saturation" exponent in equation 13.46 (only used if Wood's Law with Brie et al is requested). Switch to select between Generalized Wood's Law and Wood's Law with Brie et al in the calculation of the effective bulk modulus of the fluid
$C_{K,i}(P)$	the tabular functions of pressure in the calculation of the effective bulk modulus of the dry rock frame in equation 13.49
$C_{G,i}(P)$	the tabular functions of pressure in the calculation of the effective shear modulus of the dry rock frame in equation 13.50

Outputs

The output from the Petro-elastic Model depends on which simulated seismic-derived properties are requested (from the list of P-wave impedance, S-wave impedance and Poisson's ratio). When a simulated seismic-derived property is requested, it is output as a value on each active grid cell at each report time.

One use of these seismic-derived quantities is in history-matching using seismic data. In this application, gradients of simulated seismic-derived quantities with respect to uncertain reservoir model parameters may be required. In addition to simulated seismic-derived quantities, the Petro-elastic Model can calculate and output these gradients. For these calculations to be performed, you must specify a Petro-elastic Model

(using the keywords described in ["Summary of keywords"](#)) and request gradients. For further information on requesting gradients, see ["Gradient Option"](#).

Summary of keywords

RUNSPEC

Keyword	Description
PEDIMS	Sets the number of petro-elastic regions in the simulation, sets table dimensions, in terms of the number of rows, for the PEKTABx and PEGTABx keywords.

REGIONS section

Keyword	Description
PENUM	Specifies the petro-elastic region number for each grid cell in the simulation model.

PROPS section

Keyword	Description
PECOEFS	Specifies a variety of fluid and rock properties, and also options to be used in the petro-elastic calculations, for each petro-elastic region in the simulation model.
PEKTABx	Specifies the frame bulk modulus functions for each petro-elastic region in the simulation model.
PEGTABx	Specifies the frame shear modulus functions for each petro-elastic region in the simulation model.

SOLUTION/SUMMARY section

Keyword	Description
RPTRST	Controls output of properties (including simulated seismic-derived properties) to RESTART files. The Petro-elastic Model specific arguments for this keyword are shown below.
GRADRESV	Specifies which gradients of solution data are output to the RESTART files. The Petro-elastic Model specific arguments for this keyword are shown in below.

Note: For each of these properties, you must also request the [DEN](#) item in the [RPTRST](#) keyword.

Restriction	Keyword	Description
Eclipse 100	RPTRST	- ACIP Output pressure-wave impedance - ACIS Output shear-wave impedance - POIS Output Poisson's ratio

Restriction	Keyword	Description
Eclipse 100	GRADRESV	• ACIP Output gradient of pressure-wave impedance • ACIS Output gradient of shear-wave impedance • POIS Output gradient of Poisson's ratio